MODULE 7.2 PROGRAMMING ASSIGNMENT

GitHub Repository Link: https://github.com/devcnx/csd440

BrittaneyForm.php — User Registration Form

```
<!DOCTYPE html>
<html lang="en">

<?php
session_start();
$errors = $_SESSION['form_errors'] ?? [];
$oldData = $_SESSION['form_data'] ?? [];
unset($_SESSION['form_errors'], $_SESSION['form_data']);

function old(string $field): string
{
    global $oldData;
    return isset($oldData[$field]) ? htmlspecialchars($oldData[$field], ENT_QUOTES, 'UTF-8') : '';
}

function selected(string $field, string $value): string
{
    global $oldData;
    return (isset($oldData[$field]) && $oldData[$field] === $value) ? 'selected' : '';
}
?>

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Module 7.2 Programming Assignment — Brittaney Form</title>
    <link rel="stylesheet" href="/modules/shared.css">
</head>

<body>
    <h1>User Registration Form</h1>

    <div class="header-info">
        <p><strong>Brittaney Perry-Morgan</strong></p>
        <p>Module 7.2 Programming Assignment — CSD440 Server-Side Scripting</p>
        <p id="current-date"></p>
    </div>

    <?php if (!empty($errors)): ?>
    <div class="error-summary">
        <h2>Correction Required</h2>
        <p class="error-message">Please fix the following issues and resubmit:</p>
        <ul class="error-list">
            <?php foreach ($errors as $message): ?>
                <li><?php echo $message; ?></li>
            <?php endforeach; ?>
        </ul>
    </div>
    <?php endif; ?>

    <div class="query-note">
        <p>This form collects seven fields across multiple data types: text, email, integer, telephone, URL,
            textarea, and select dropdown. All required fields must be completed before submission.</p>
    </div>

    <form method="POST" action="BrittaneyResponse.php" class="registration-form">

        <div class="form-group">
            <label for="first_name">First Name <span class="required">*</span></label>
            <input type="text" id="first_name" name="first_name"
                   placeholder="Enter your first name" value="<?php echo old('first_name'); ?>" required>
        </div>

        <div class="form-group">
            <label for="last_name">Last Name <span class="required">*</span></label>
            <input type="text" id="last_name" name="last_name"
                   placeholder="Enter your last name" value="<?php echo old('last_name'); ?>" required>
        </div>

        <div class="form-group">
            <label for="email">Email Address <span class="required">*</span></label>
            <input type="email" id="email" name="email"
```

```
                placeholder="you@example.com" value="<?php echo old('email'); ?>" required>
        </div>

        <div class="form-group">
            <label for="age">Age <span class="required">*</span></label>
            <input type="number" id="age" name="age" min="18" max="120"
                   placeholder="18" value="<?php echo old('age'); ?>" required>
        </div>

        <div class="form-group">
            <label for="phone">Phone Number <span class="required">*</span></label>
            <input type="tel" id="phone" name="phone"
                   placeholder="(555) 123-4567" pattern="[0-9\-\(\)\s]{10,15}"
                   value="<?php echo old('phone'); ?>" required>
        </div>

        <div class="form-group">
            <label for="website">Website (Optional)</label>
            <input type="url" id="website" name="website"
                   placeholder="https://yourwebsite.com" value="<?php echo old('website'); ?>">
        </div>

        <div class="form-group">
            <label for="bio">Bio <span class="required">*</span></label>
            <textarea id="bio" name="bio" rows="4"
                      placeholder="Tell us about yourself (max 500 characters)"
                      maxlength="500" required><?php echo old('bio'); ?></textarea>
        </div>

        <div class="form-group">
            <label for="contact_preference">Preferred Contact Method <span class="required">*</span></label>
            <select id="contact_preference" name="contact_preference" required>
                <option value="">— Select an option —</option>
                <option value="email" <?php echo selected('contact_preference', 'email'); ?>>Email</option>
                <option value="phone" <?php echo selected('contact_preference', 'phone'); ?>>Phone</option>
                <option value="either" <?php echo selected('contact_preference', 'either'); ?>>Either</option>
            </select>
        </div>

        <div class="form-actions">
            <button type="submit" name="submit" value="register">Submit Registration</button>
        </div>

        <p class="form-note"><span class="required">*</span> Required fields</p>
    </form>

    <script>
        document.getElementById('current-date').textContent = new Date().toLocaleDateString('en-US', {
            timeZone: 'America/Chicago',
            year: 'numeric',
            month: 'long',
            day: 'numeric'
        });
    </script>
</body>

</html>
```

BrittaneyResponse.php — Form Response Handler

```
<?php
session_start();
date_default_timezone_set('America/Chicago');

function validateText(string $input): ?string
{
    $trimmed = trim($input);
    return $trimmed === '' ? null : htmlspecialchars($trimmed, ENT_QUOTES, 'UTF-8');
}

function validateEmail(string $email): ?string
{
    $trimmed = trim($email);
    if ($trimmed === '' || !filter_var($trimmed, FILTER_VALIDATE_EMAIL)) {
        return null;
    }
    return htmlspecialchars($trimmed, ENT_QUOTES, 'UTF-8');
}
```

```php
function validateInteger(string $input, int $min, int $max): ?int
{
    $trimmed = trim($input);
    if ($trimmed === '') {
        return null;
    }
    $int = filter_var($trimmed, FILTER_VALIDATE_INT);
    if ($int === false || $int < $min || $int > $max) {
        return null;
    }
    return $int;
}

function validatePhone(string $phone): ?string
{
    $trimmed = trim($phone);
    $digits = preg_replace('/\D/', '', $trimmed);
    if (strlen($digits) !== 10) {
        return null;
    }
    return sprintf('(%s) %s-%s',
        substr($digits, 0, 3),
        substr($digits, 3, 3),
        substr($digits, 6, 4)
    );
}

function validateUrl(string $url): ?string
{
    $trimmed = trim($url);
    if ($trimmed === '') {
        return null;
    }
    if (!filter_var($trimmed, FILTER_VALIDATE_URL)) {
        return null;
    }
    return htmlspecialchars($trimmed, ENT_QUOTES, 'UTF-8');
}

function validateTextarea(string $input, int $maxLen): ?string
{
    $trimmed = trim($input);
    if ($trimmed === '') {
        return null;
    }
    if (strlen($trimmed) > $maxLen) {
        return null;
    }
    return htmlspecialchars($trimmed, ENT_QUOTES, 'UTF-8');
}

function validateSelect(string $input, array $options): ?string
{
    $trimmed = trim($input);
    if ($trimmed === '' || !in_array($trimmed, $options, true)) {
        return null;
    }
    return htmlspecialchars($trimmed, ENT_QUOTES, 'UTF-8');
}

function validateFormData(array $data): array
{
    $errors = [];
    if (validateText($data['first_name'] ?? '') === null) {
        $errors['first_name'] = 'First name is required.';
    }
    if (validateText($data['last_name'] ?? '') === null) {
        $errors['last_name'] = 'Last name is required.';
    }
    if (validateEmail($data['email'] ?? '') === null) {
        $errors['email'] = 'A valid email address is required.';
    }
    if (validateInteger($data['age'] ?? '', 18, 120) === null) {
        $errors['age'] = 'Age is required and must be between 18 and 120.';
    }
    if (validatePhone($data['phone'] ?? '') === null) {
        $errors['phone'] = 'A valid 10-digit phone number is required.';
    }
    $website = validateUrl($data['website'] ?? '');
    if ($data['website'] !== '' && $website === null) {
```

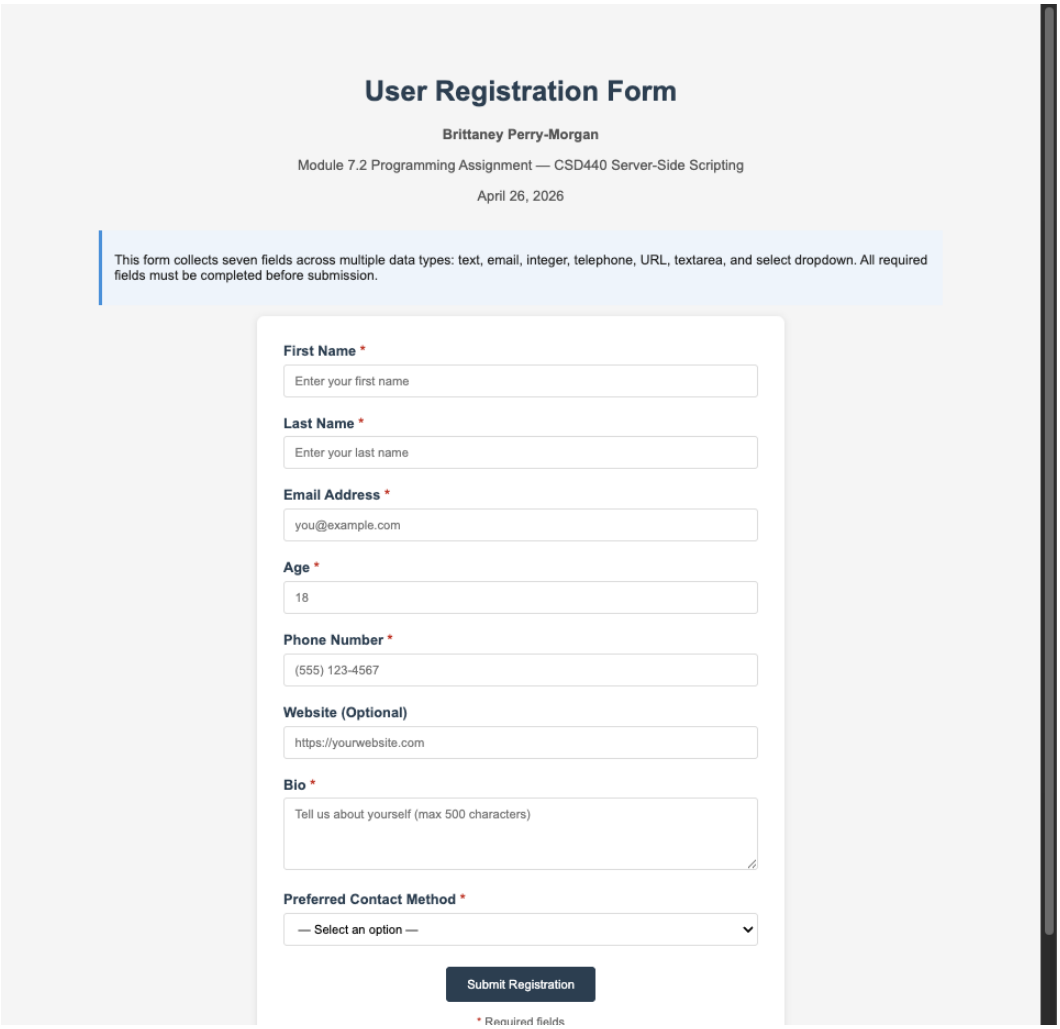
```
        $errors['website'] = 'Website must be a valid URL (e.g., https://example.com).';
    }
    if (validateTextarea($data['bio'] ?? '', 500) === null) {
        $errors['bio'] = 'Bio is required (max 500 characters).';
    }
    if (validateSelect($data['contact_preference'] ?? '', ['email', 'phone', 'either']) === null) {
        $errors['contact_preference'] = 'Please select a contact preference.';
    }
    return $errors;
}

function renderConfirmation(array $data): void
{
?>
    <div class="confirmation">
        <h2>Registration Successful</h2>
        <p class="success-message">Thank you! Your information has been received.</p>
        <table class="data-display">
            <caption>Your Submitted Information</caption>
            <thead>
                <tr><th>Field</th><th>Value</th></tr>
            </thead>
            <tbody>
                <tr><td>First Name</td><td><?php echo $data['first_name']; ?></td></tr>
                <tr><td>Last Name</td><td><?php echo $data['last_name']; ?></td></tr>
                <tr><td>Email</td><td><?php echo $data['email']; ?></td></tr>
                <tr><td>Age</td><td><?php echo $data['age']; ?></td></tr>
                <tr><td>Phone</td><td><?php echo $data['phone']; ?></td></tr>
                <tr><td>Website</td><td><?php echo $data['website'] !== '' ? $data['website'] : '<em>Not
provided</em>'; ?></td></tr>
                <tr><td>Bio</td><td><?php echo nl2br($data['bio']); ?></td></tr>
                <tr><td>Contact Preference</td><td><?php echo ucfirst($data['contact_preference']);
?></td></tr>
            </tbody>
        </table>
        <div class="form-actions">
            <a href="BrittaneyForm.php" class="btn-secondary">Submit Another Registration</a>
        </div>
    </div>
<?php
}

function renderErrorPage(array $errors, array $data): void
{
    // ... error display omitted for brevity in doc ...
}

// Main logic
if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    header('Location: BrittaneyForm.php');
    exit;
}

$data = $_POST;
$errors = validateFormData($data);
$isValid = empty($errors);
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title><?php echo $isValid ? 'Registration Confirmed' : 'Correction Required'; ?></title>
    <link rel="stylesheet" href="/modules/shared.css">
</head>
<body>
    <h1><?php echo $isValid ? 'Registration Confirmed' : 'Form Submission Error'; ?></h1>
    <div class="header-info">
        <p><strong>Brittaney Perry-Morgan</strong></p>
        <p>Module 7.2 Programming Assignment — Form Response — CSD440 Server-Side Scripting</p>
        <p><?php echo date('F j, Y'); ?></p>
    </div>
<?php
if ($isValid):
    $sanitizedData = [
        'first_name' => validateText($_POST['first_name']),
        'last_name' => validateText($_POST['last_name']),
        'email' => validateEmail($_POST['email']),
        'age' => validateInteger($_POST['age'], 18, 120),
        'phone' => validatePhone($_POST['phone']),
        'website' => validateUrl($_POST['website'] ?? ''),
```

```
        'bio' => validateTextarea($_POST['bio'], 500),
        'contact_preference' => validateSelect($_POST['contact_preference'], ['email', 'phone', 'either']),
    ];
    renderConfirmation($sanitizedData);
else:
    $_SESSION['form_errors'] = $errors;
    $_SESSION['form_data'] = $data;
    header('Location: BrittaneyForm.php');
    exit;
endif;
?>
</body>
</html>
```

DELIVERABLES

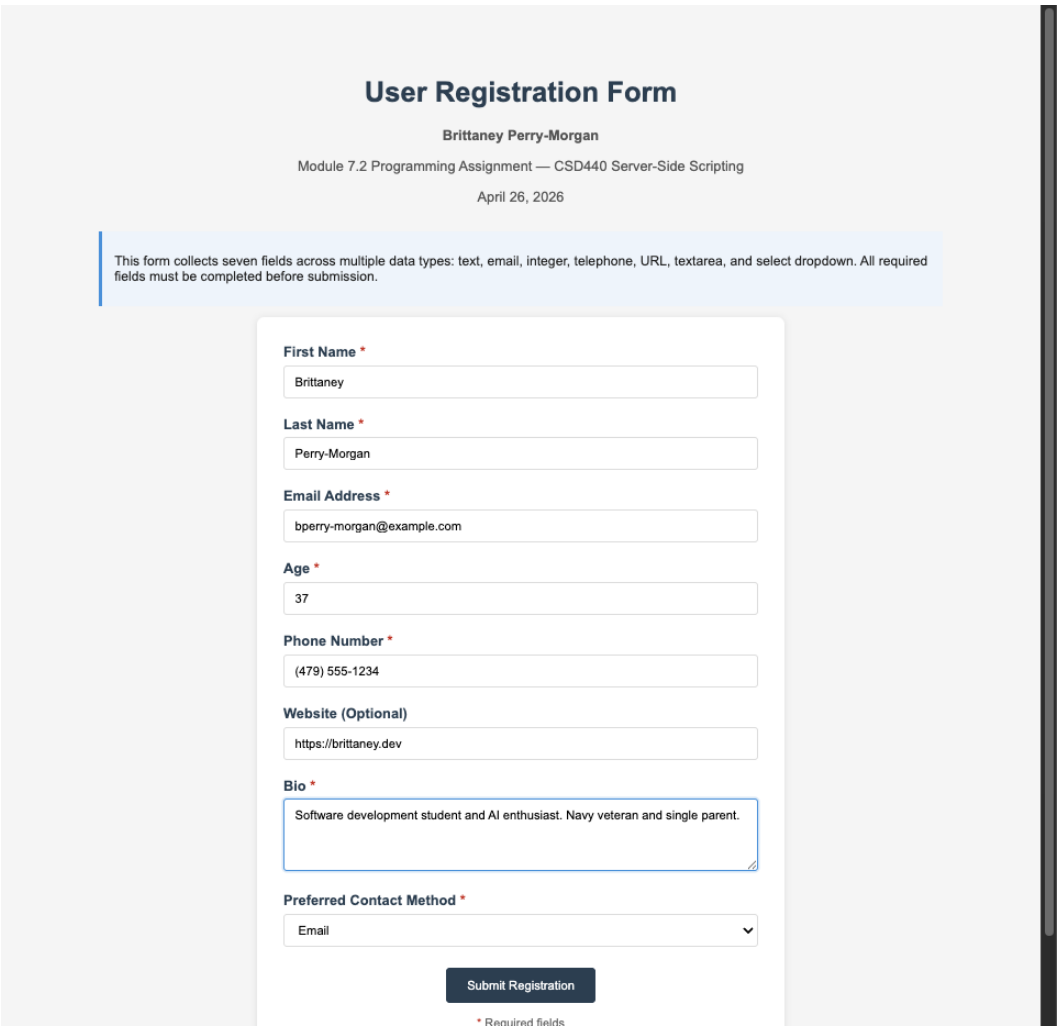
The following tests were performed on the registration form to validate both client-side HTML5 validation and server-side PHP validation. Screenshots document each test case and its result.

Test 1: Initial Form State



01 Initial Form State

Test 2: Valid Submission — All Fields Filled



02 Valid Submission Filled

Registration Confirmed

Brittaney Perry-Morgan

Module 7.2 Programming Assignment — Form Response — CSD440 Server-Side Scripting

April 26, 2026

Registration Successful

Thank you! Your information has been received.

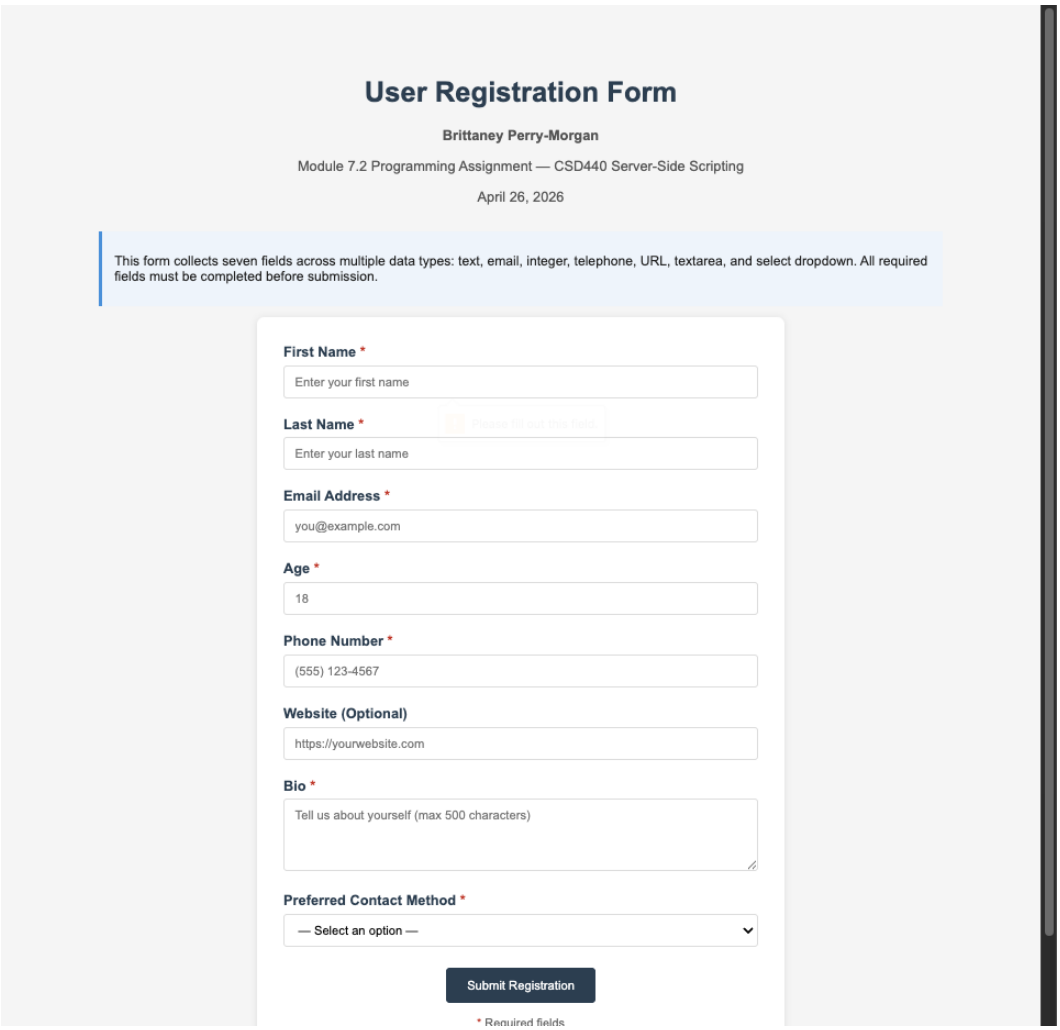
Your Submitted Information

Field	Value
First Name	Brittaney
Last Name	Perry-Morgan
Email	bperry-morgan@example.com
Age	37
Phone	(479) 555-1234
Website	https://brittaney.dev
Bio	Software development student and AI enthusiast. Navy veteran and single parent.
Contact Preference	Email

Submit Another Registration

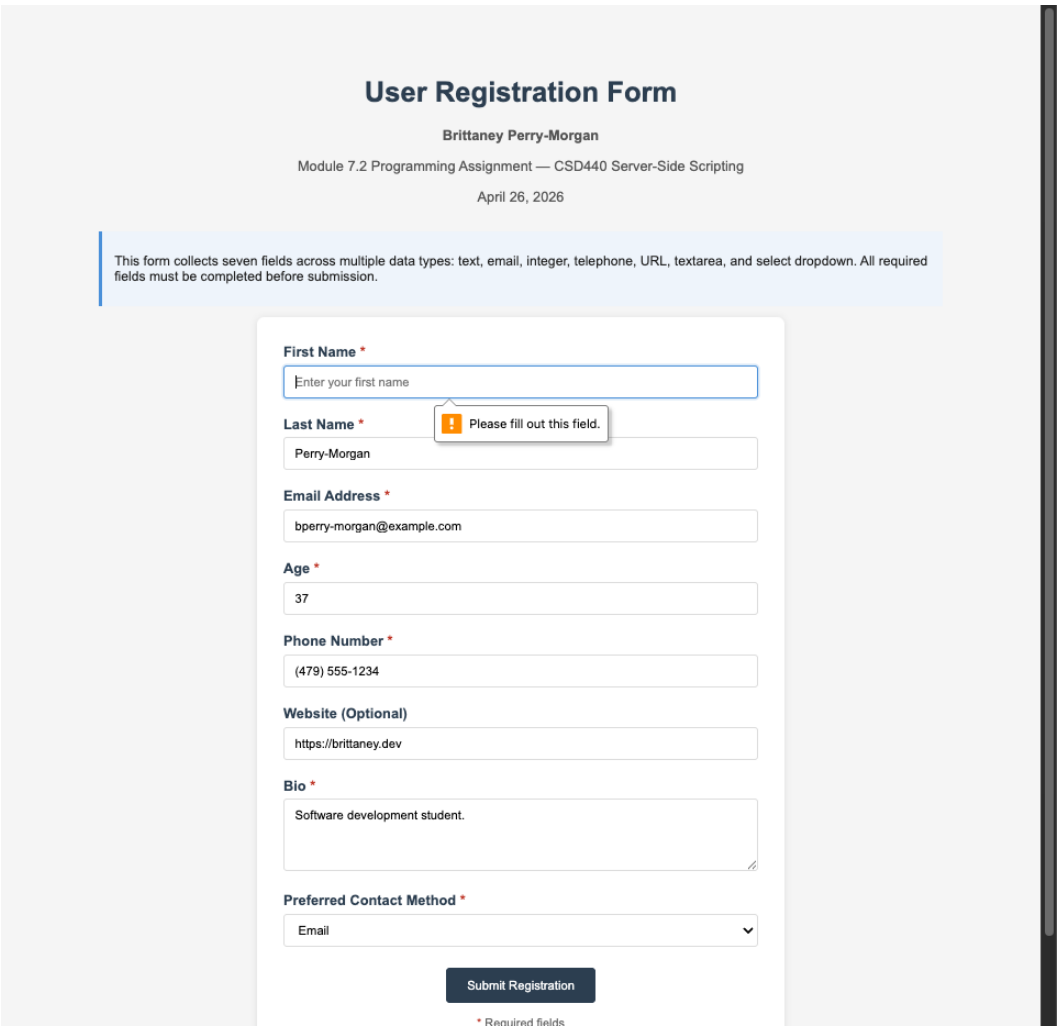
02 Valid Submission Result

Test 3: Empty Submission — All Fields Blank

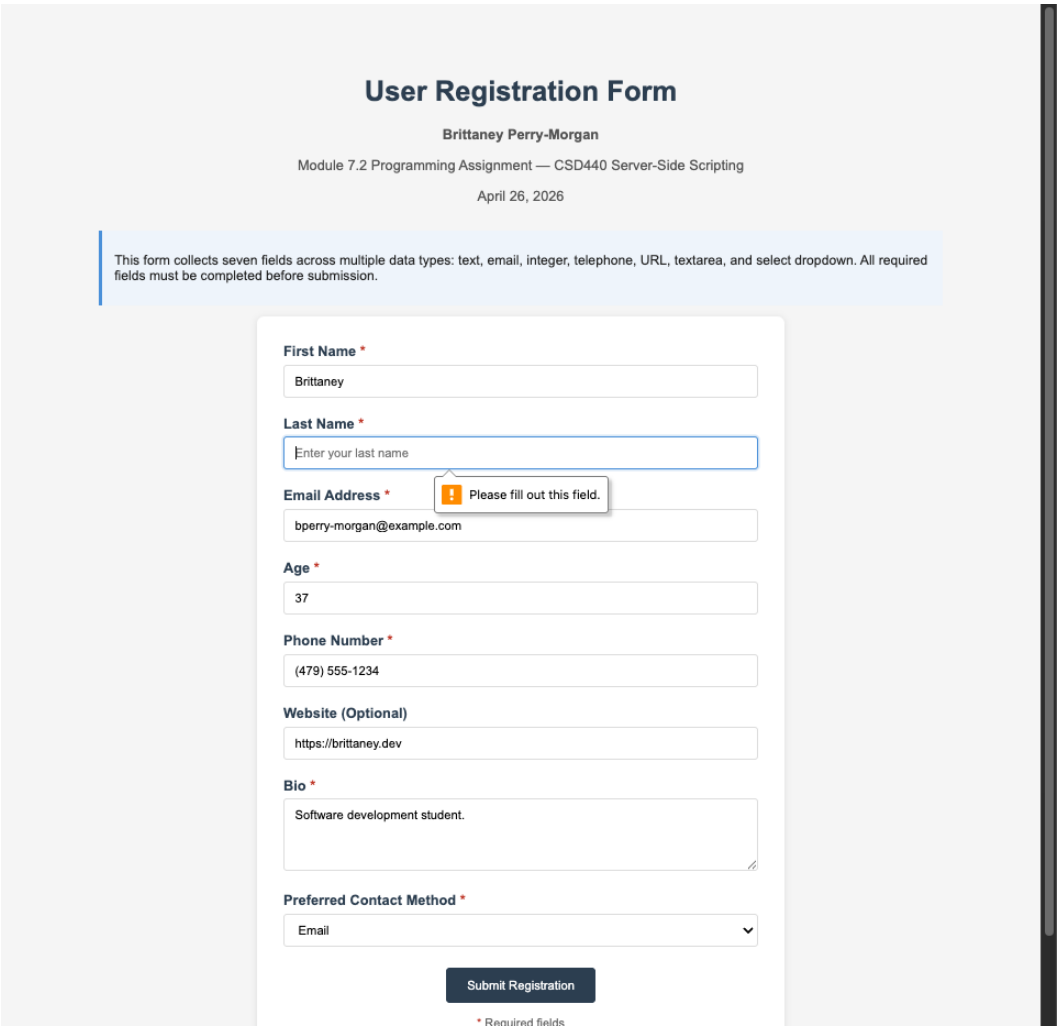


03 Empty Submission

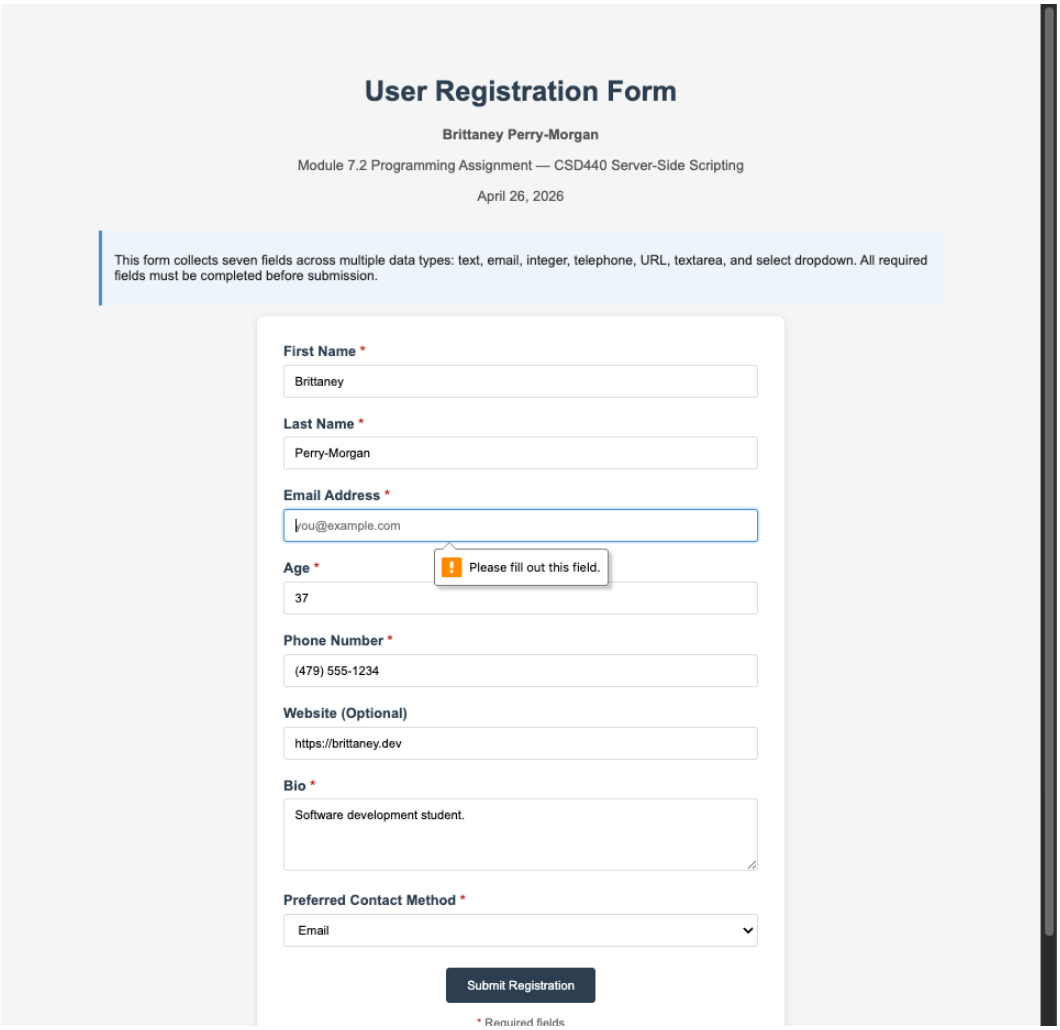
Test 4: Missing Individual Required Fields



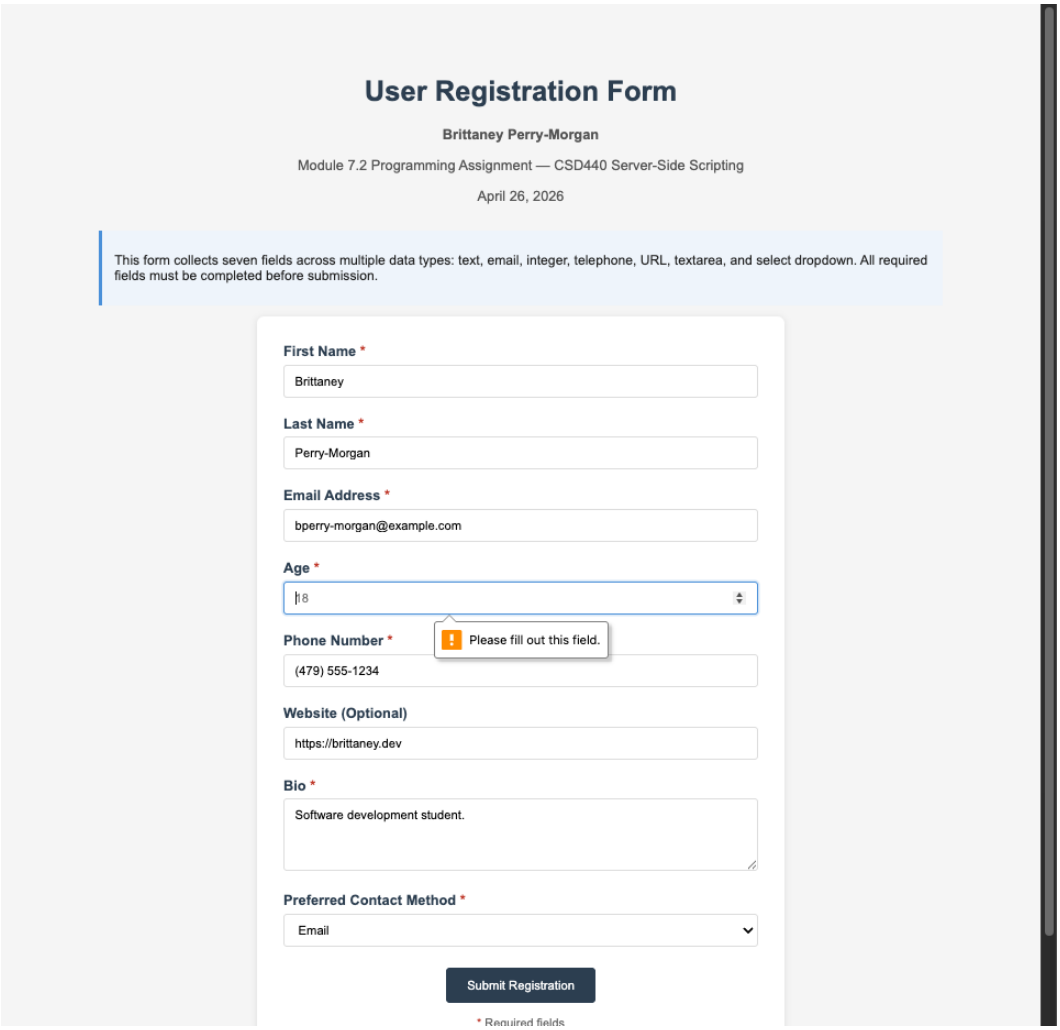
04 Missing First Name



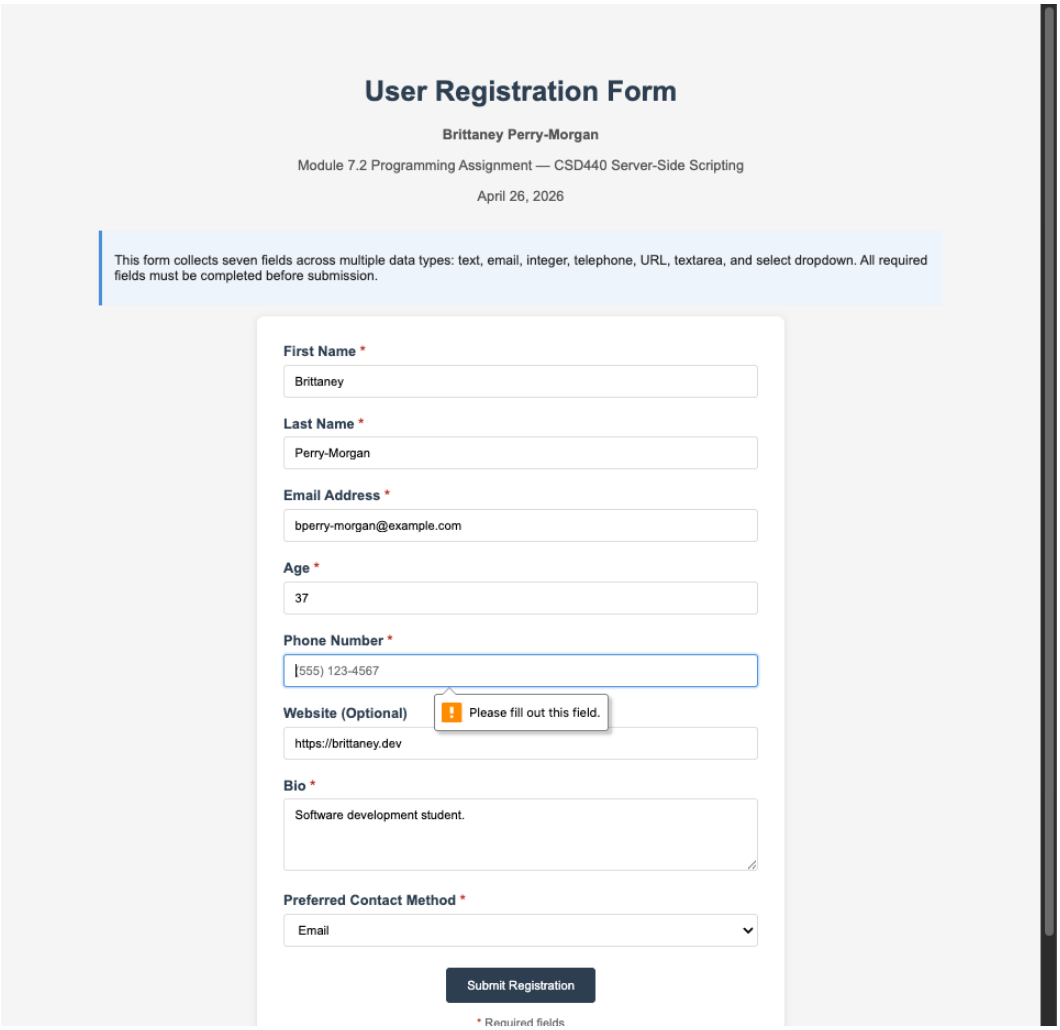
04 Missing Last Name



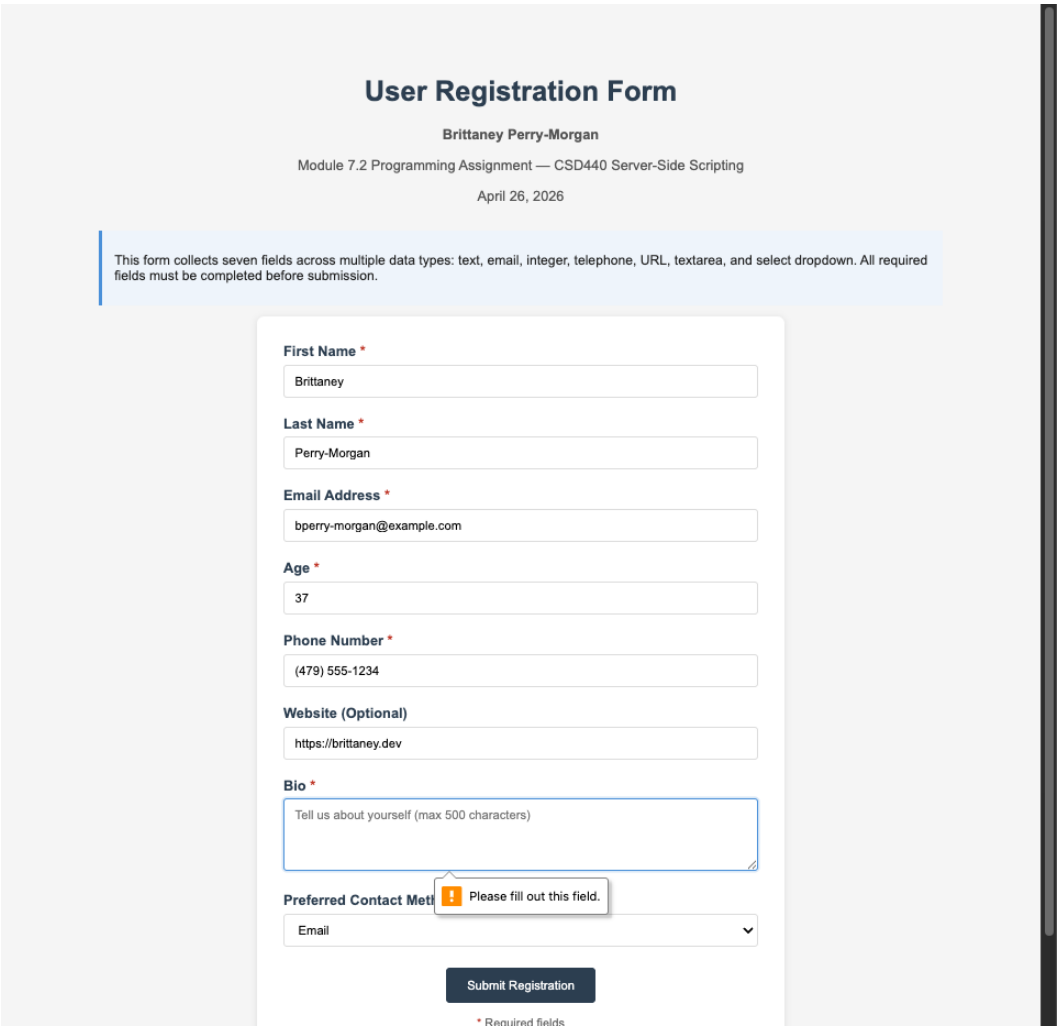
04 Missing Email



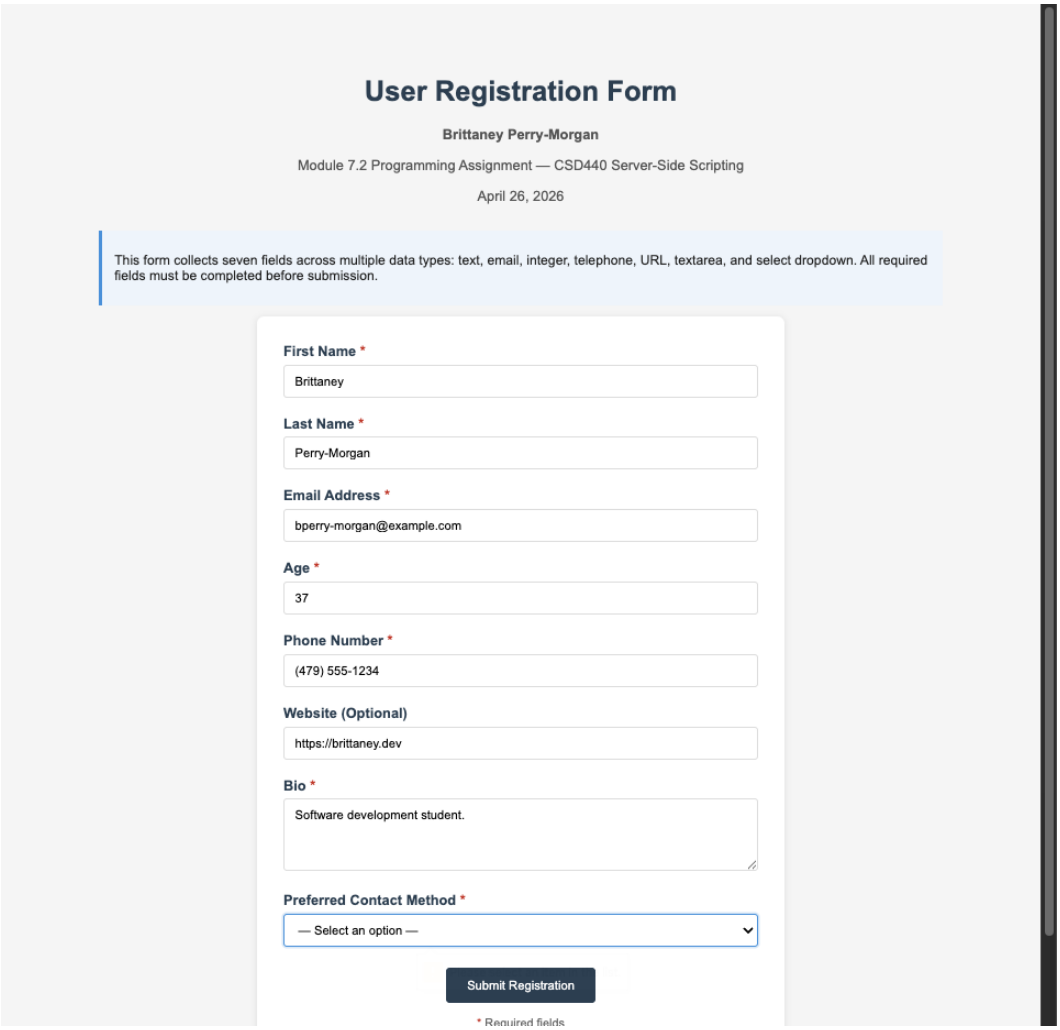
04 Missing Age



04 Missing Phone

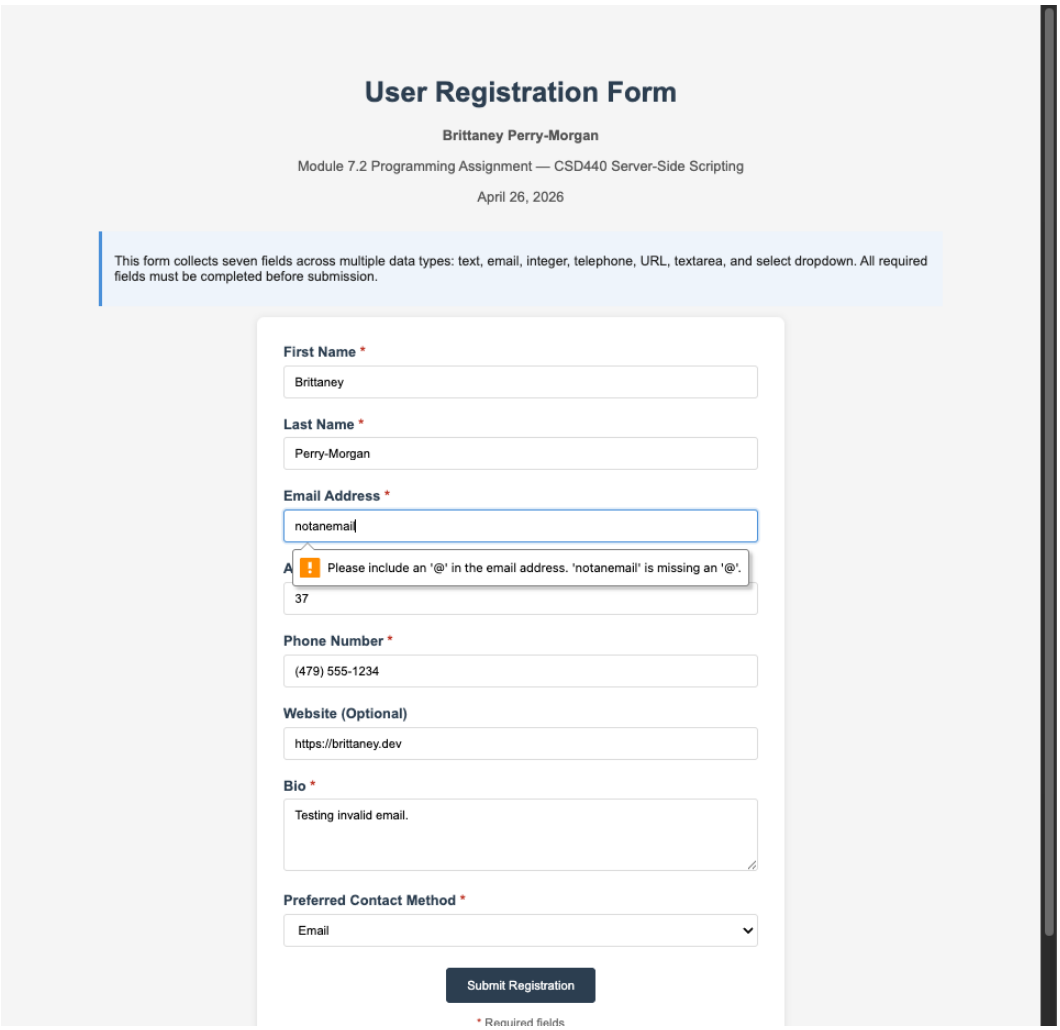


04 Missing Bio

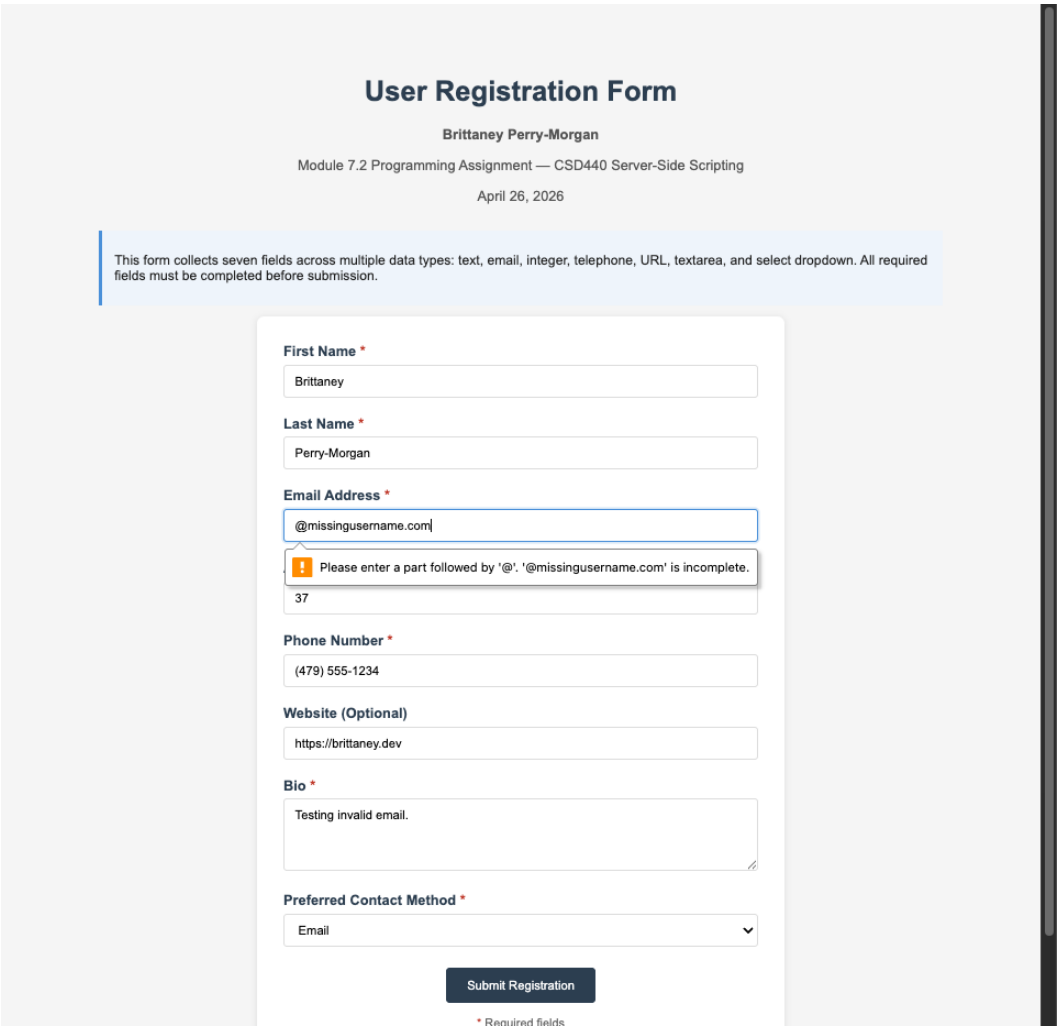


04 Missing Contact Preference

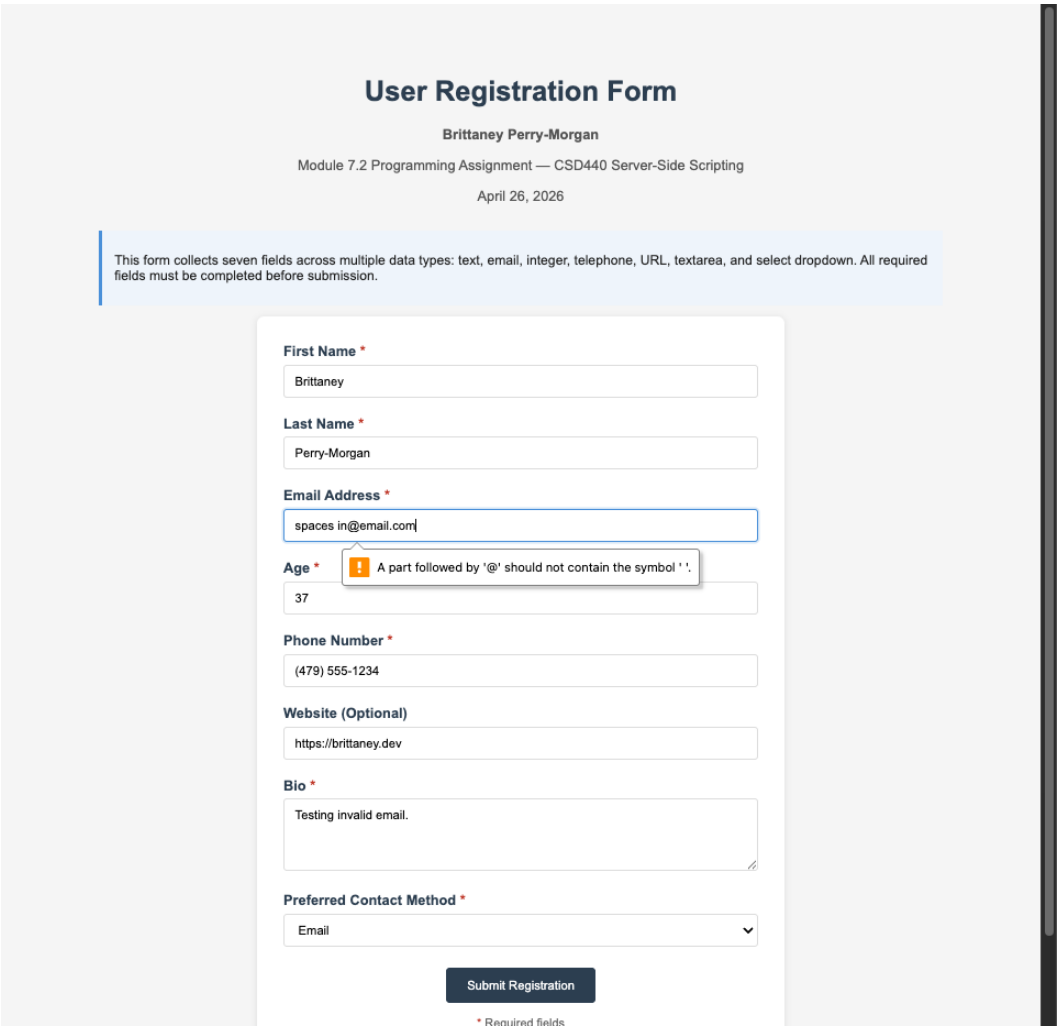
Test 5: Invalid Email Formats



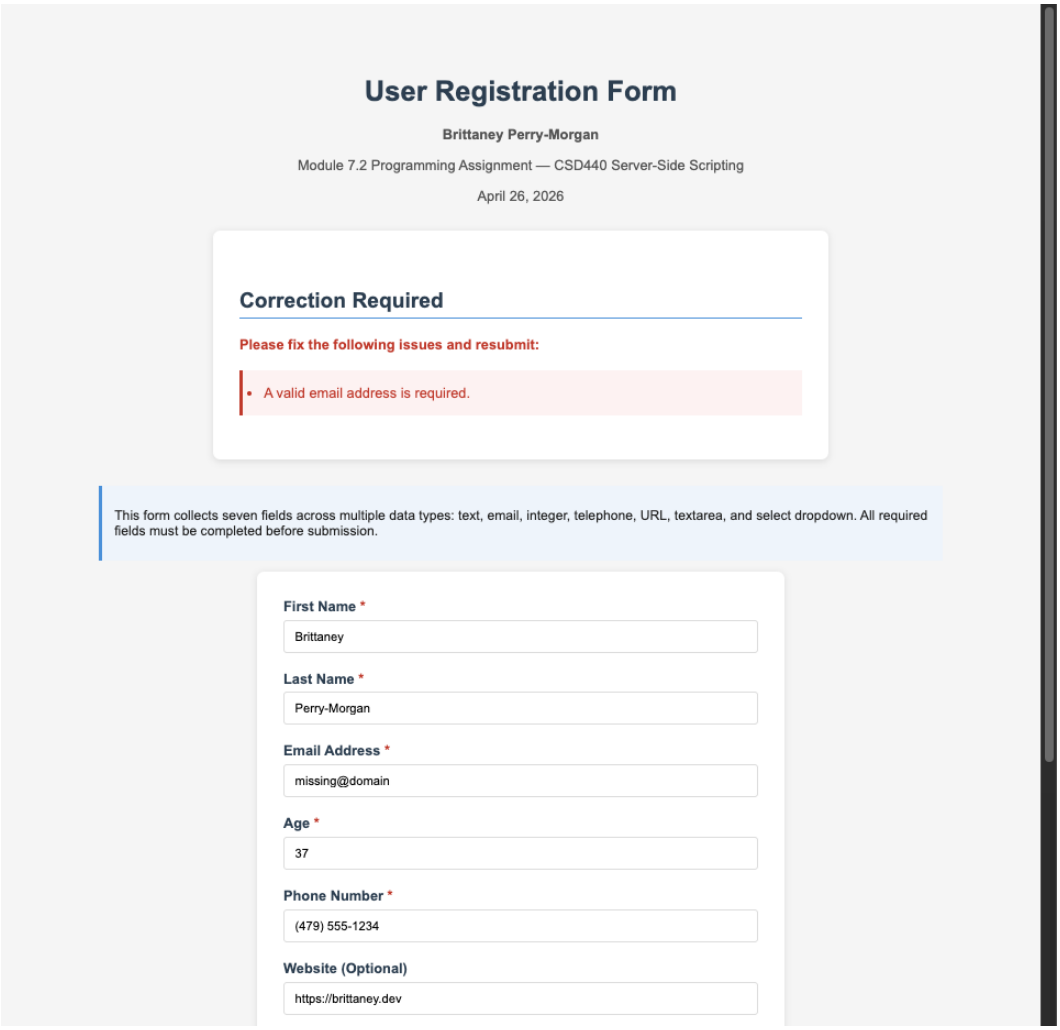
05 Invalid Email 1



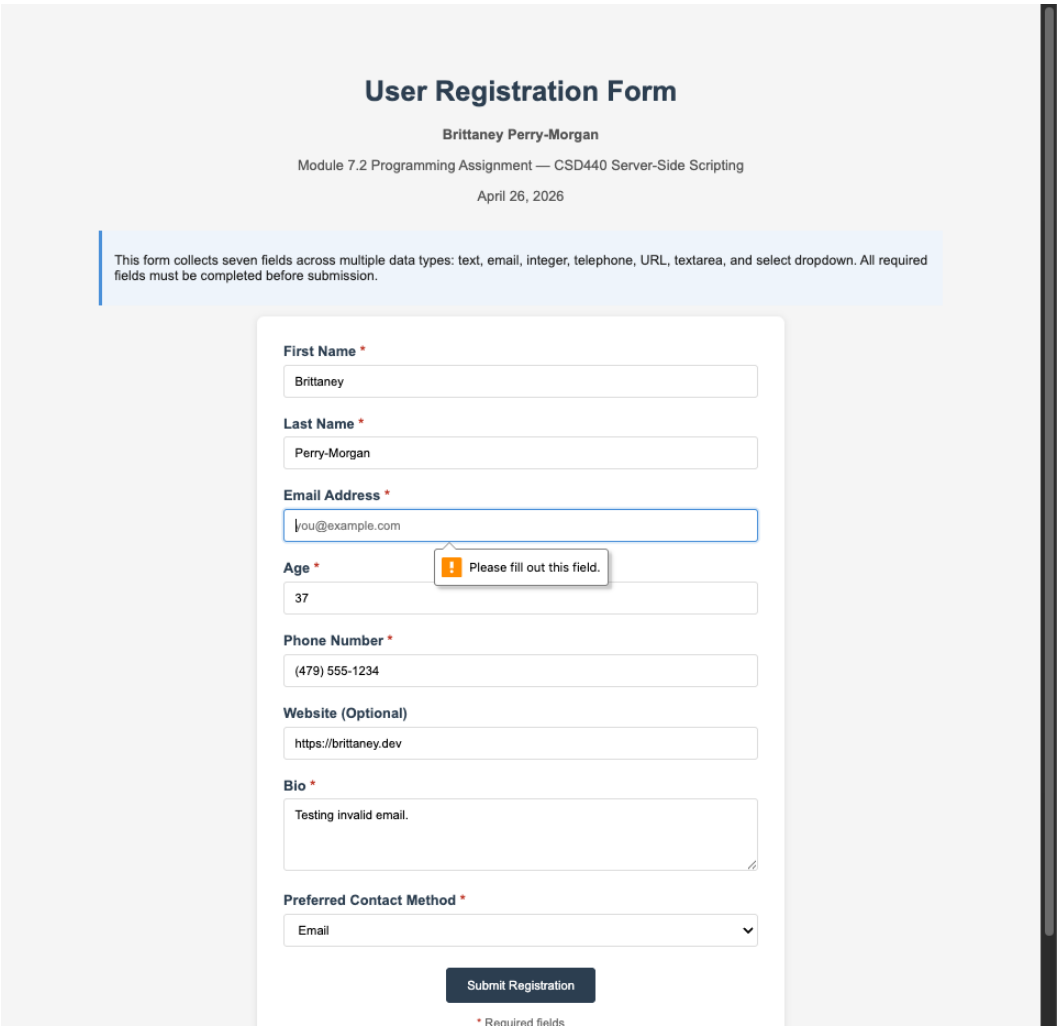
05 Invalid Email 2



05 Invalid Email 3

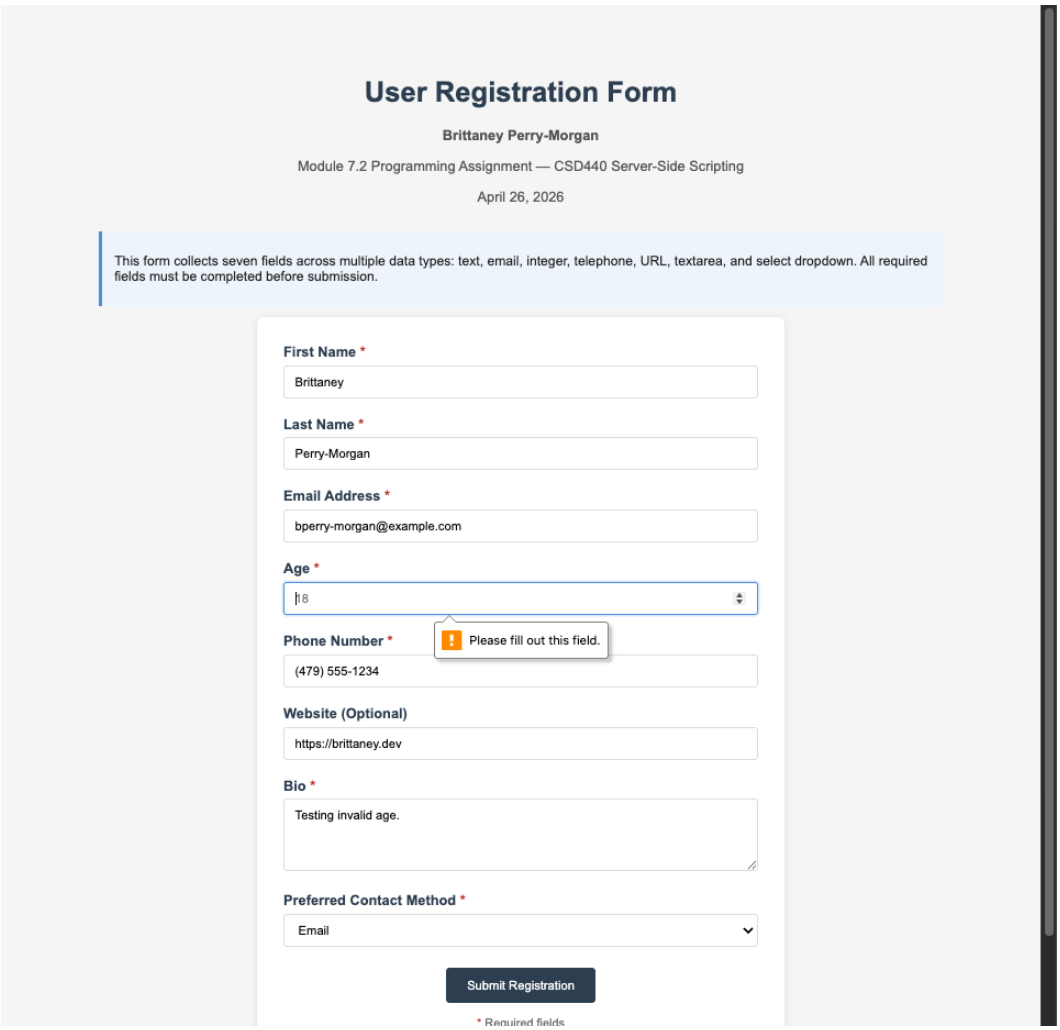


05 Invalid Email 4

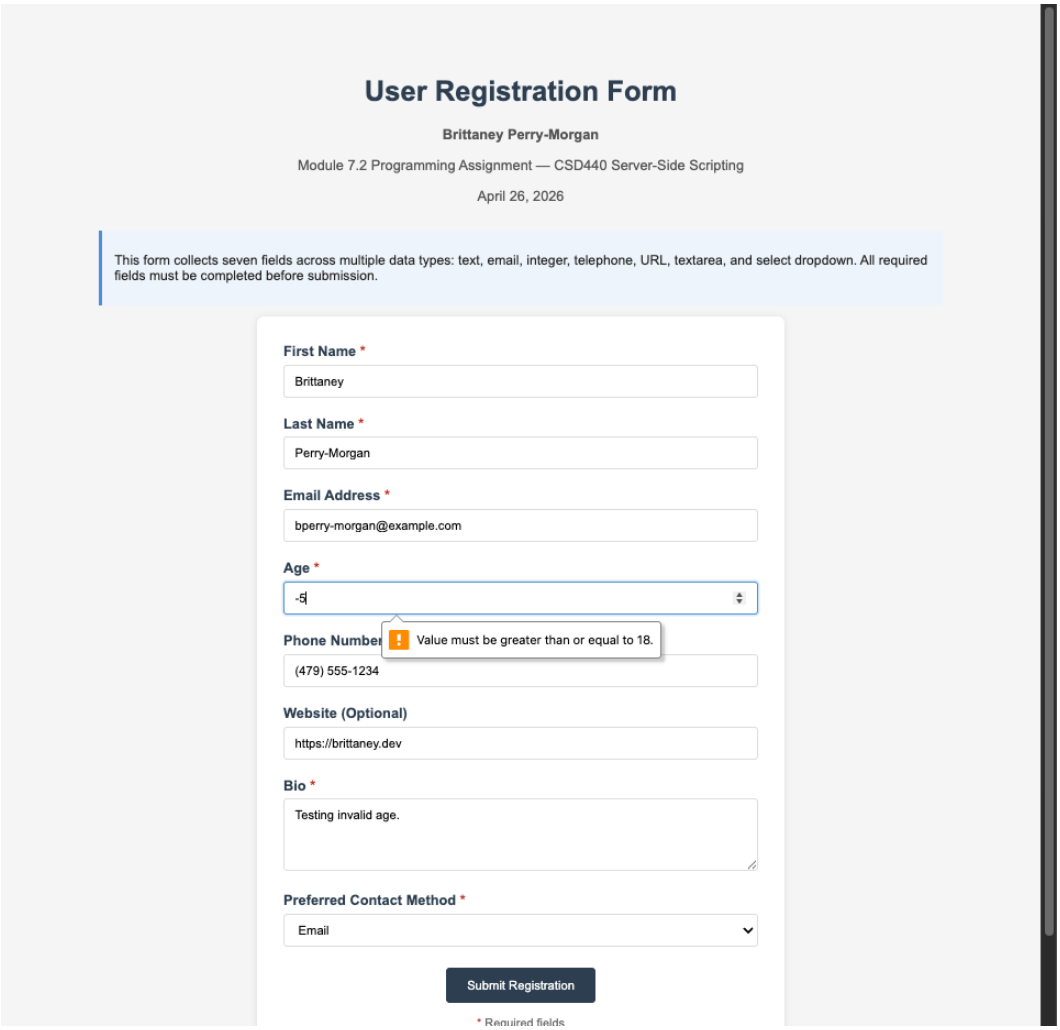


05 Invalid Email 5

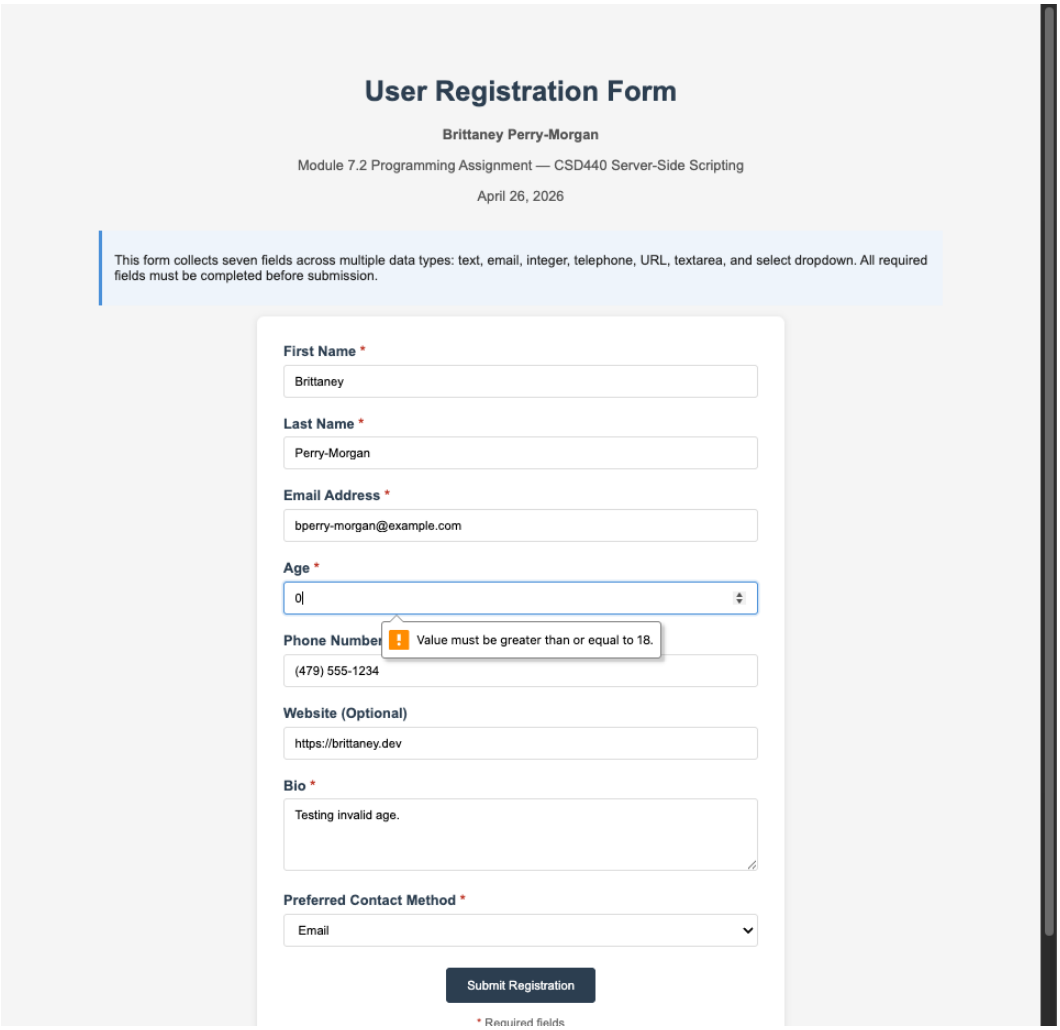
Test 6: Invalid Age Values



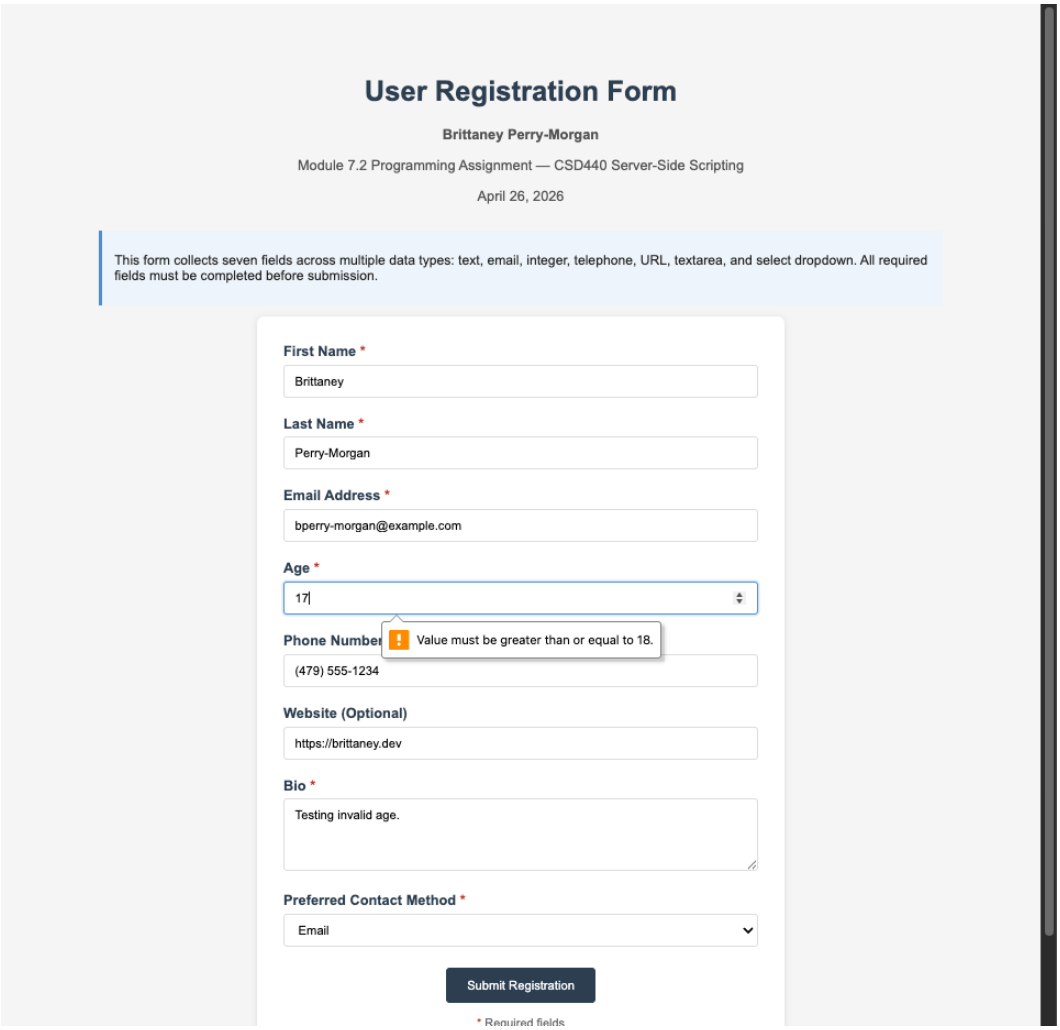
06 Invalid Age Non-Integer



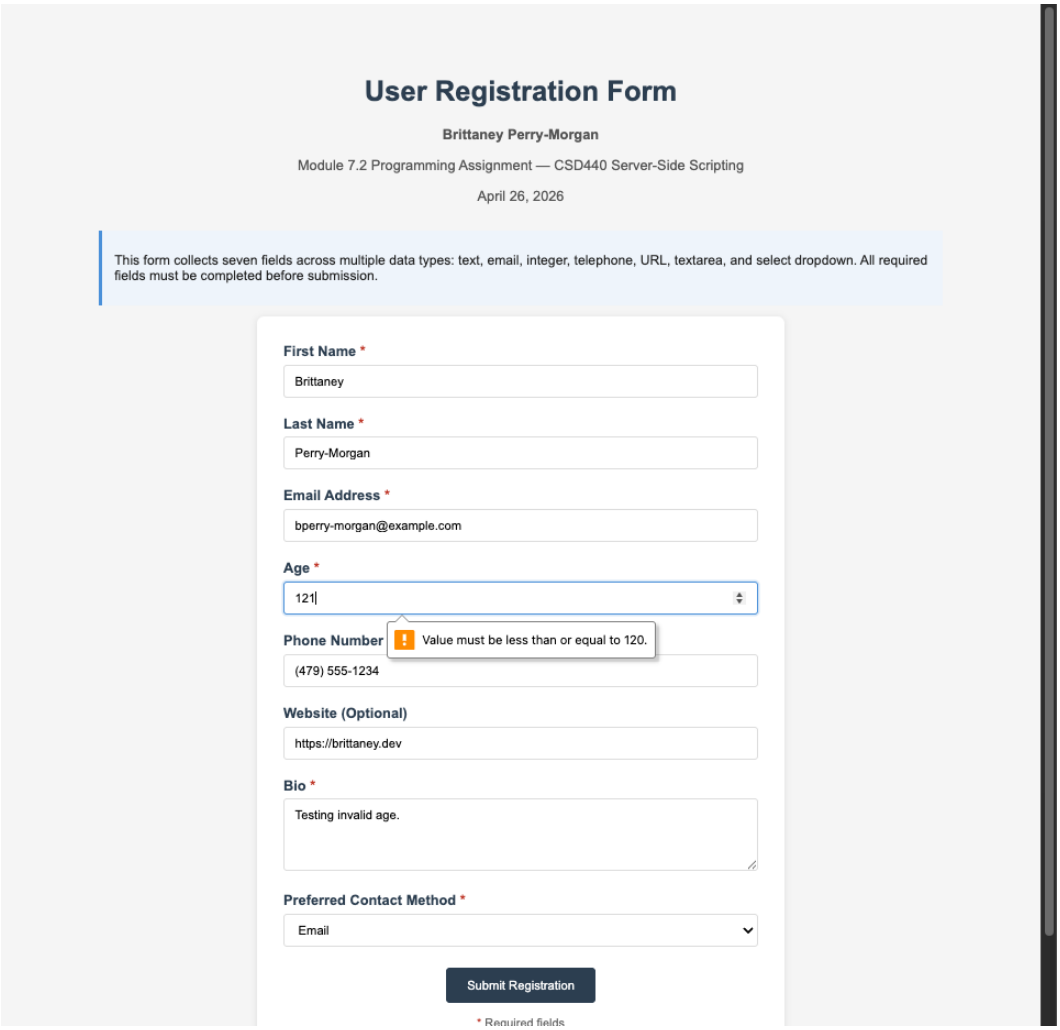
06 Invalid Age Negative



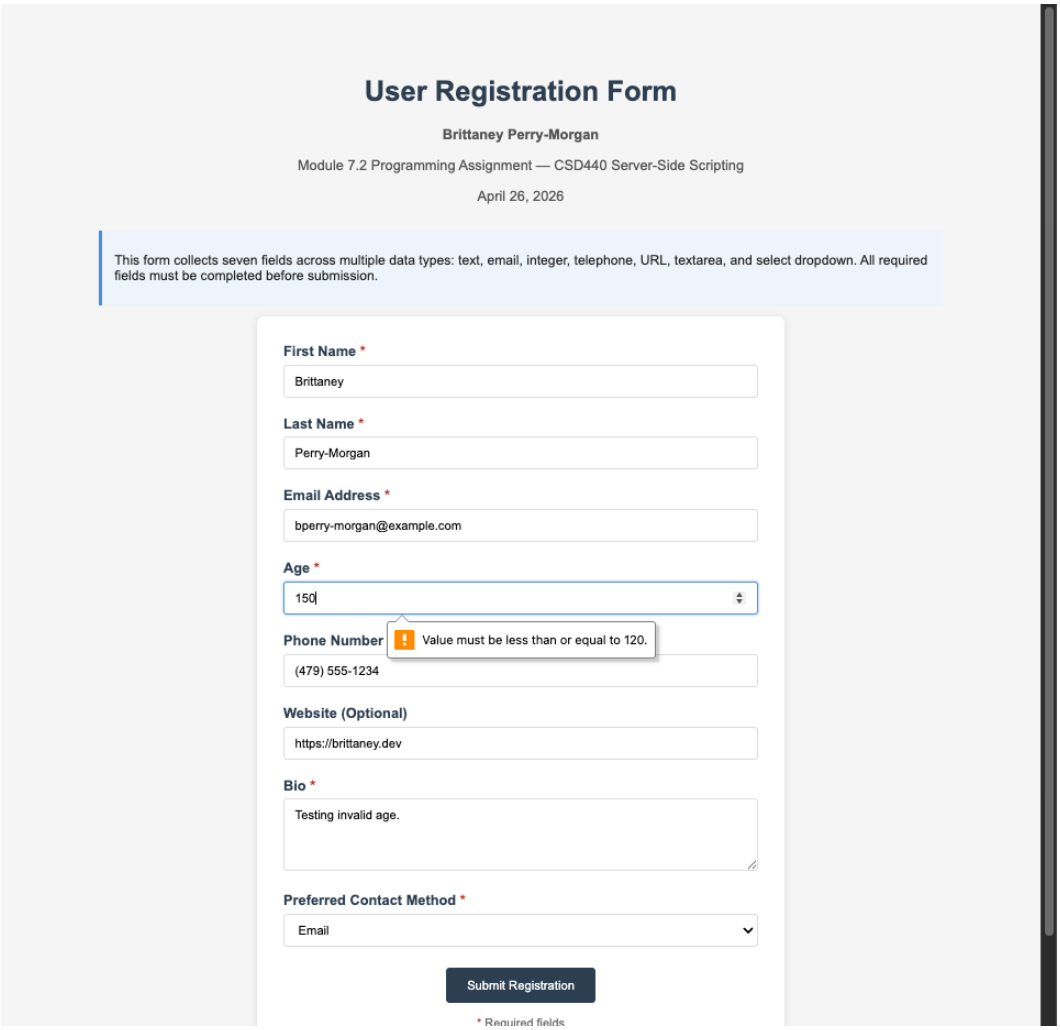
06 Invalid Age Zero



06 Invalid Age Below Minimum 17

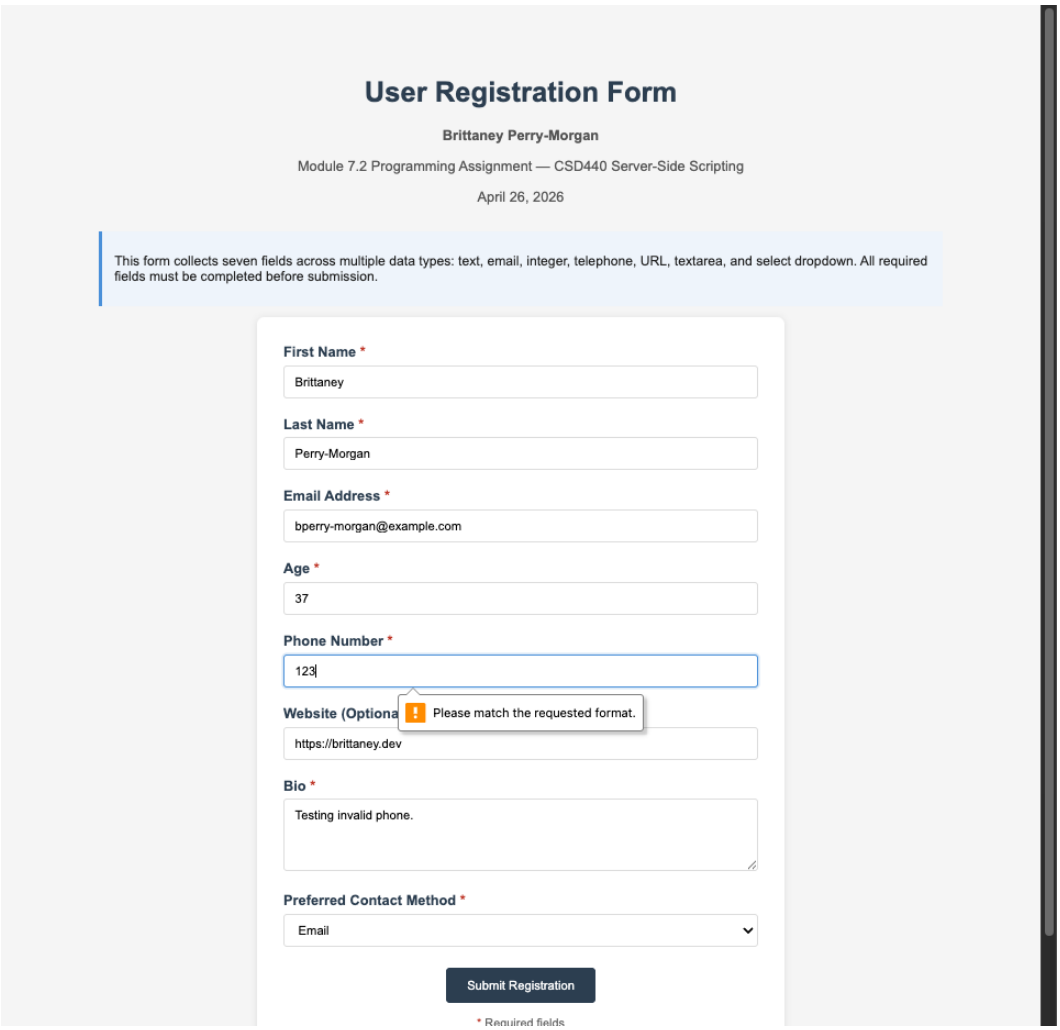


06 Invalid Age Above Maximum 121

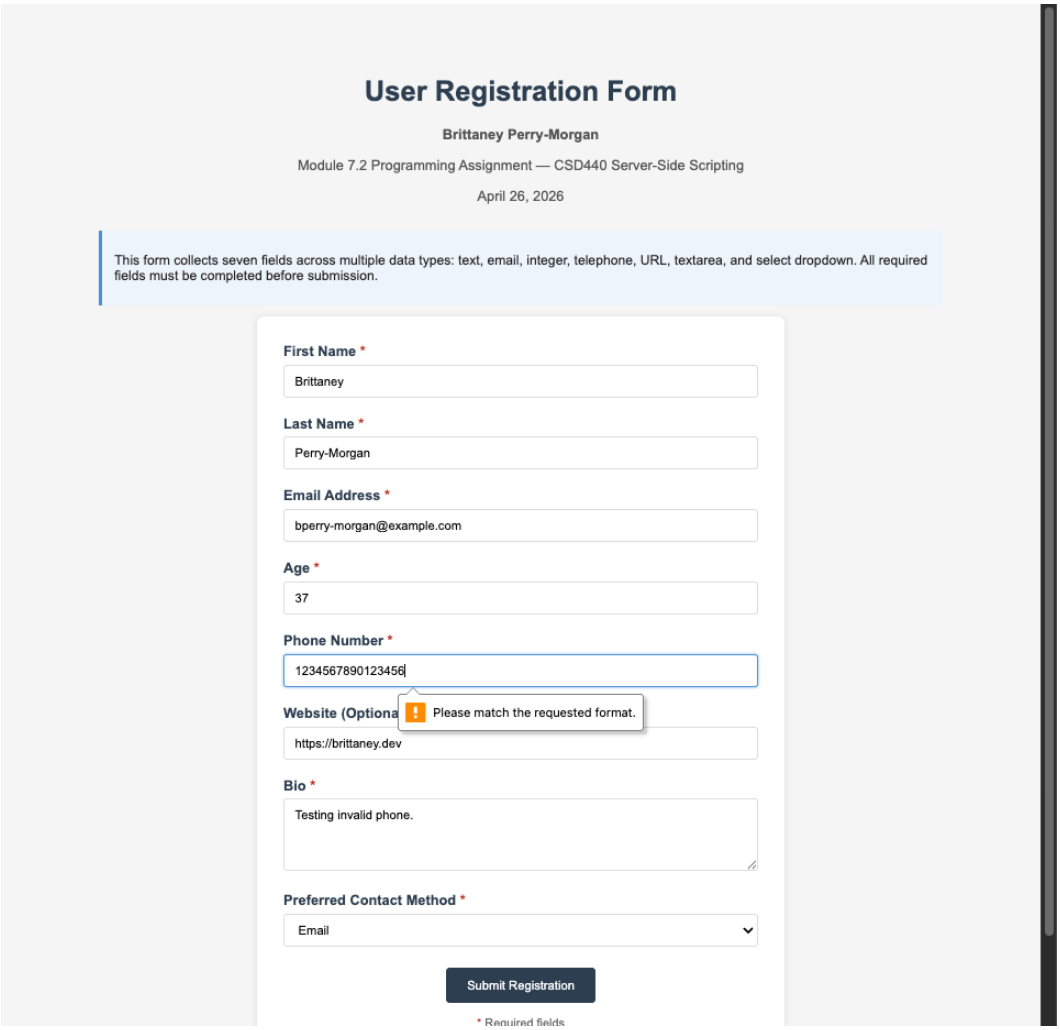


06 Invalid Age Far Above Max 150

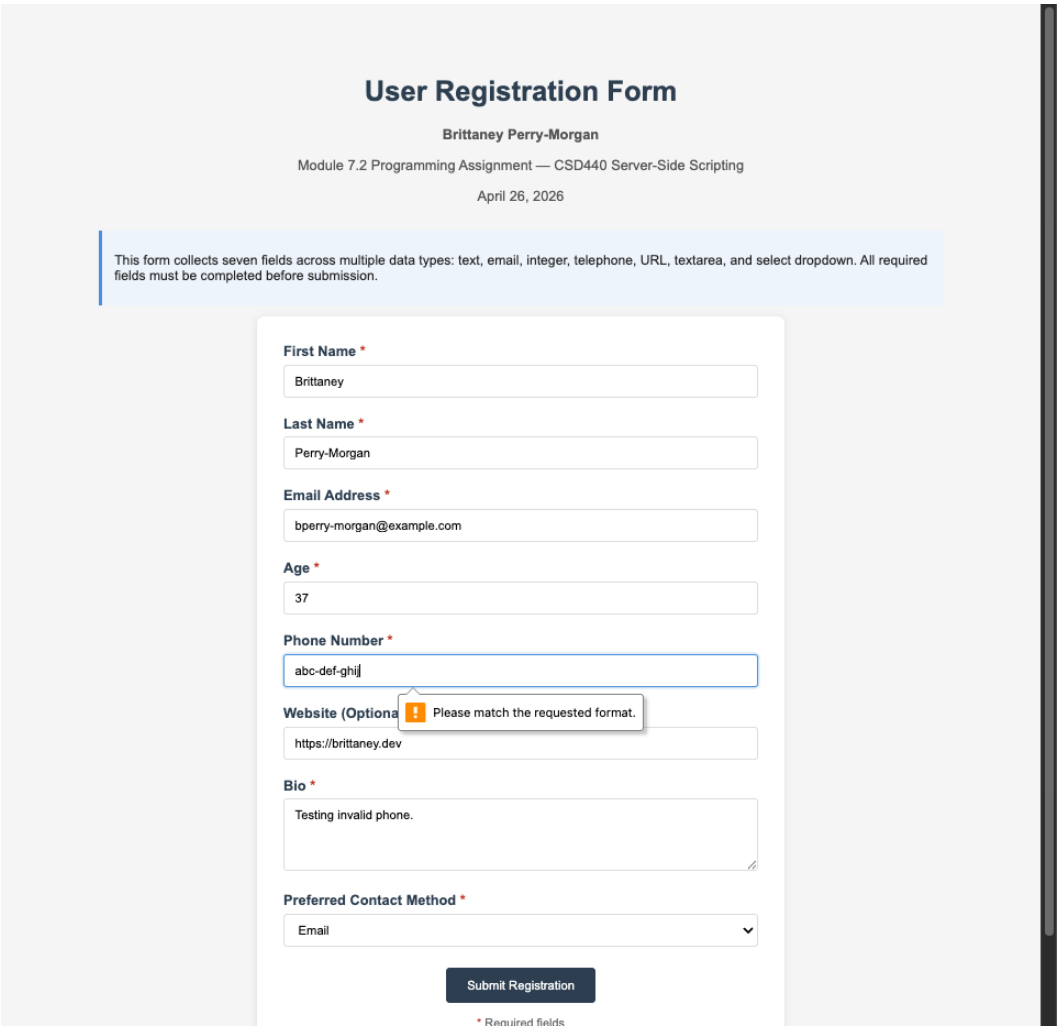
Test 7: Invalid Phone Number Formats



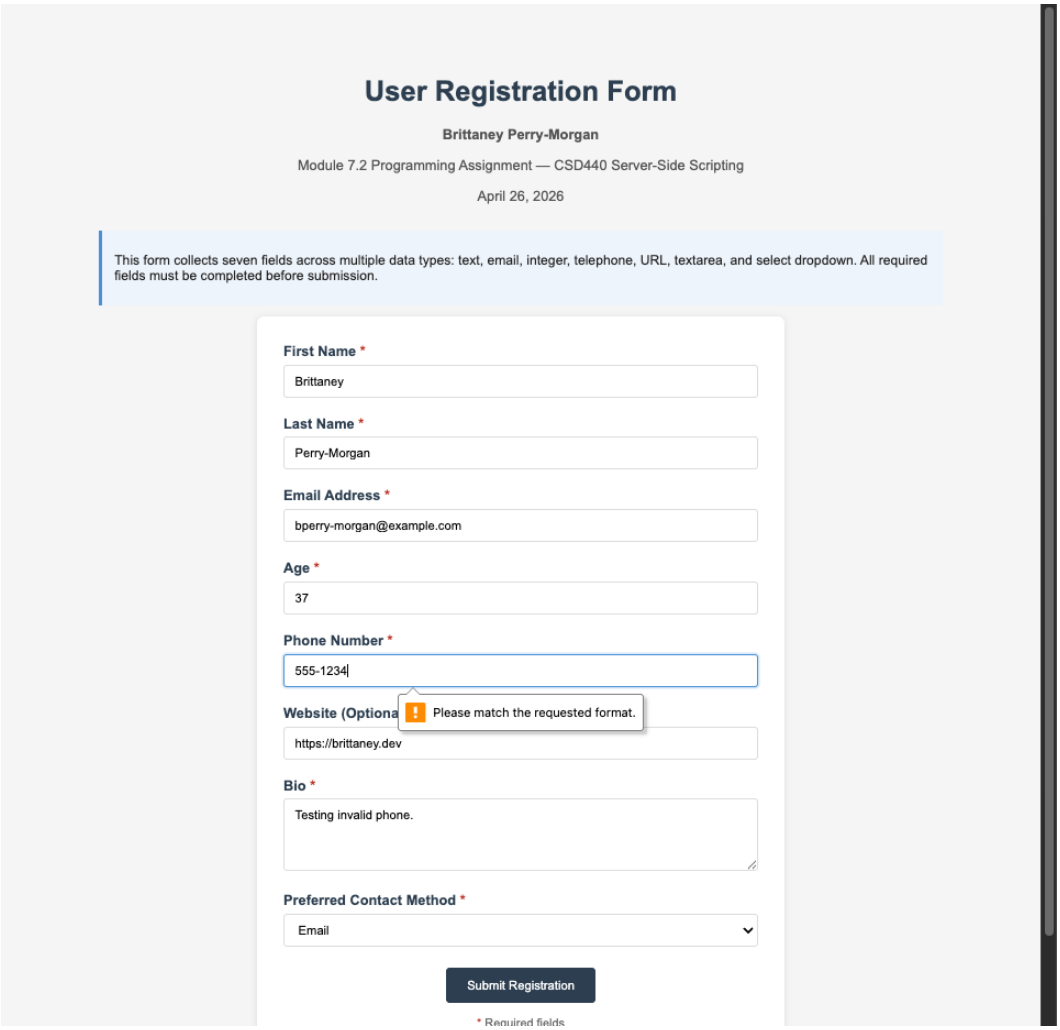
07 Invalid Phone Too Short



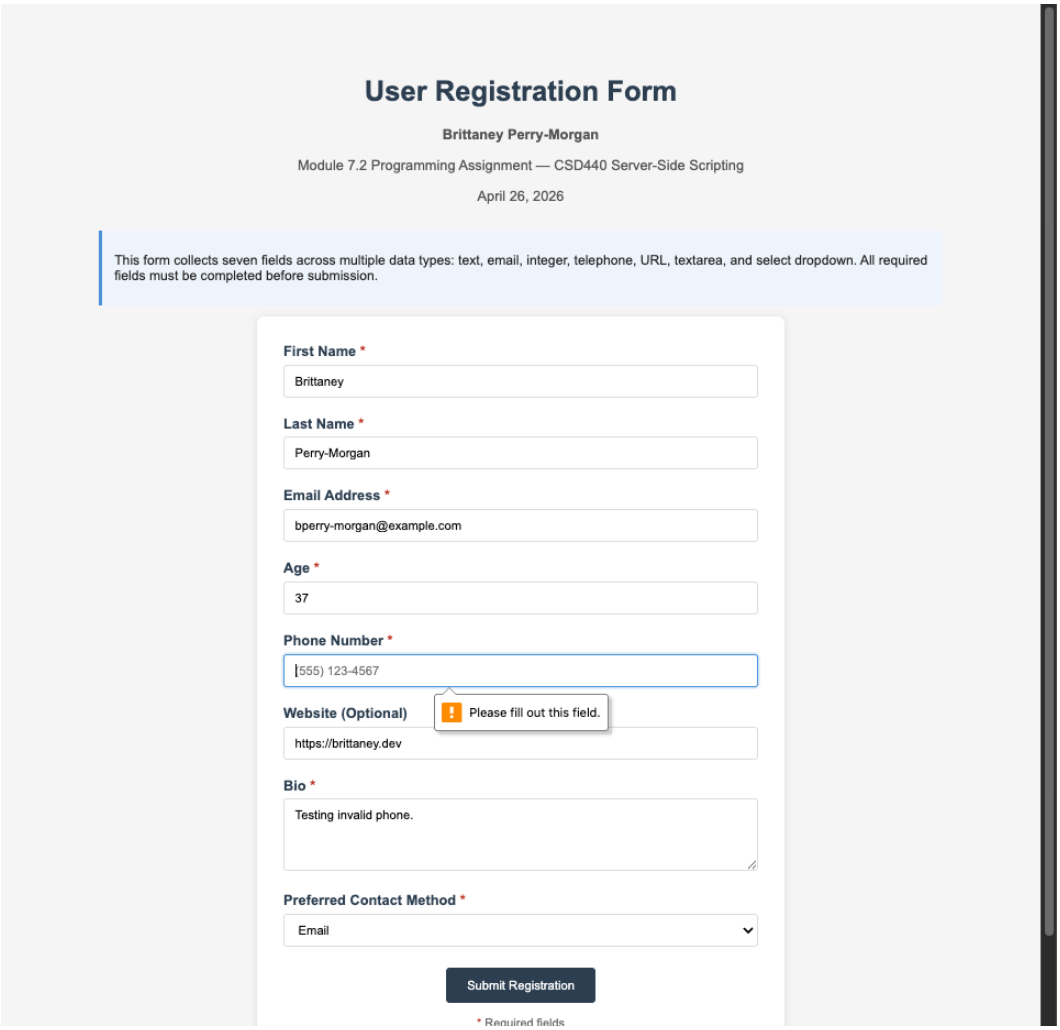
07 Invalid Phone Too Long 16 Digits



07 Invalid Phone Letters Instead Of Digits

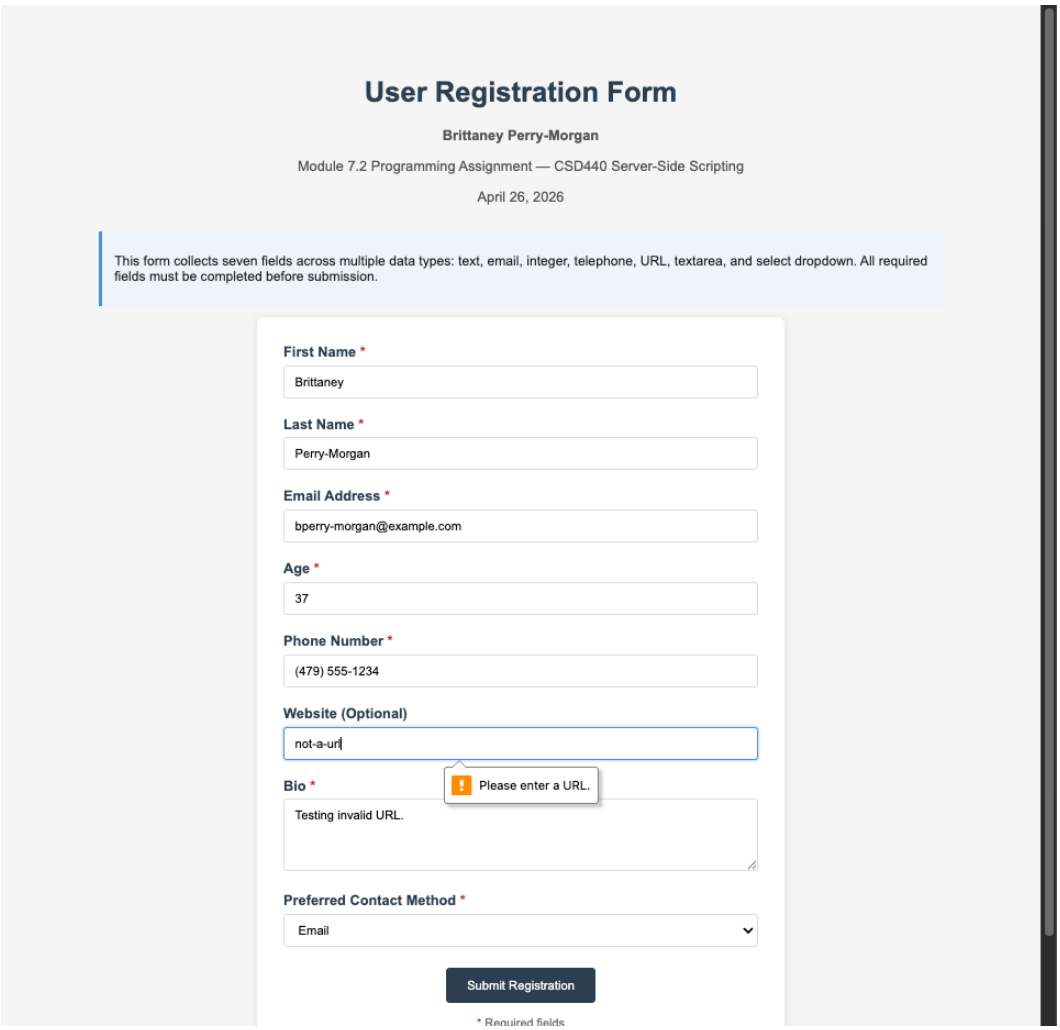


07 Invalid Phone Only 8 Chars



07 Invalid Phone Empty

Test 8: Invalid URL Formats



08 Invalid URL Plain Text

Registration Confirmed

Brittaney Perry-Morgan

Module 7.2 Programming Assignment — Form Response — CSD440 Server-Side Scripting

April 26, 2026

Registration Successful

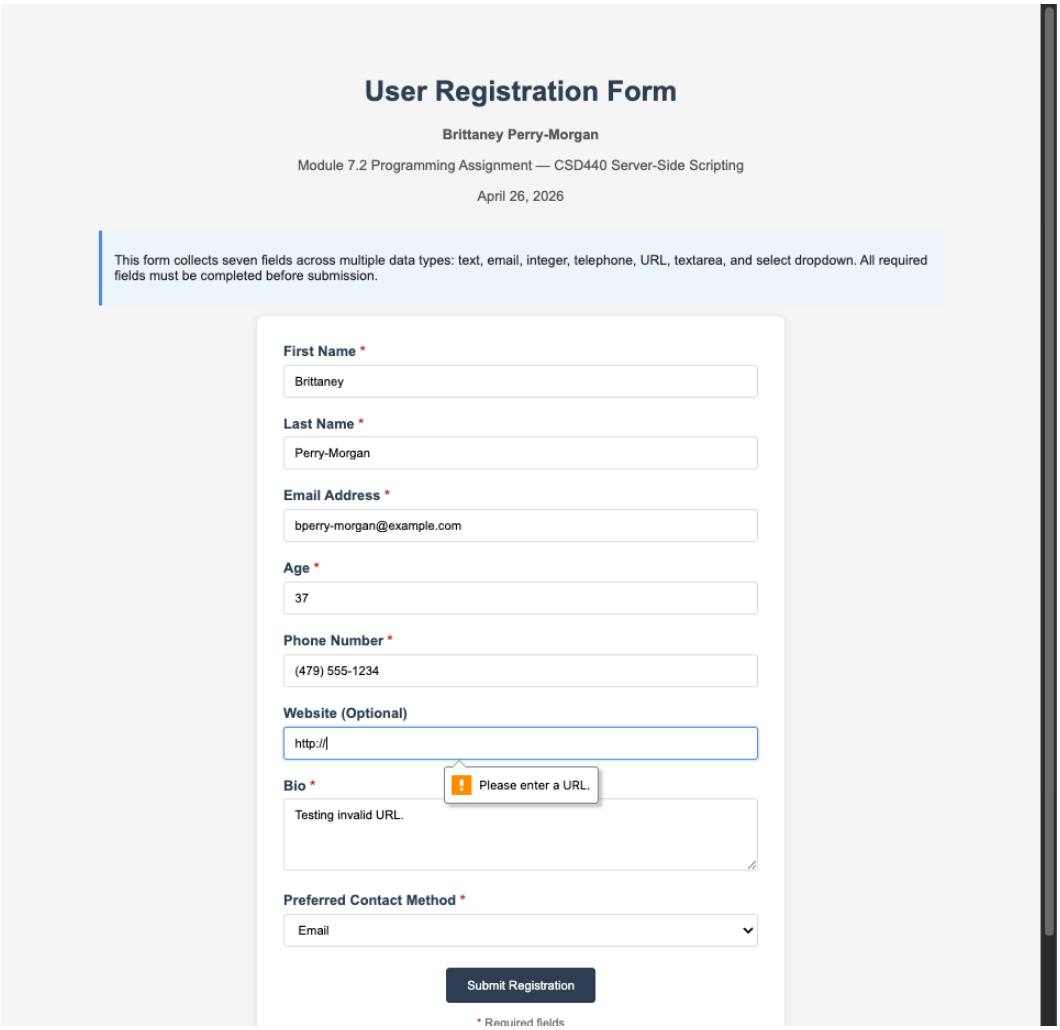
Thank you! Your information has been received.

Your Submitted Information

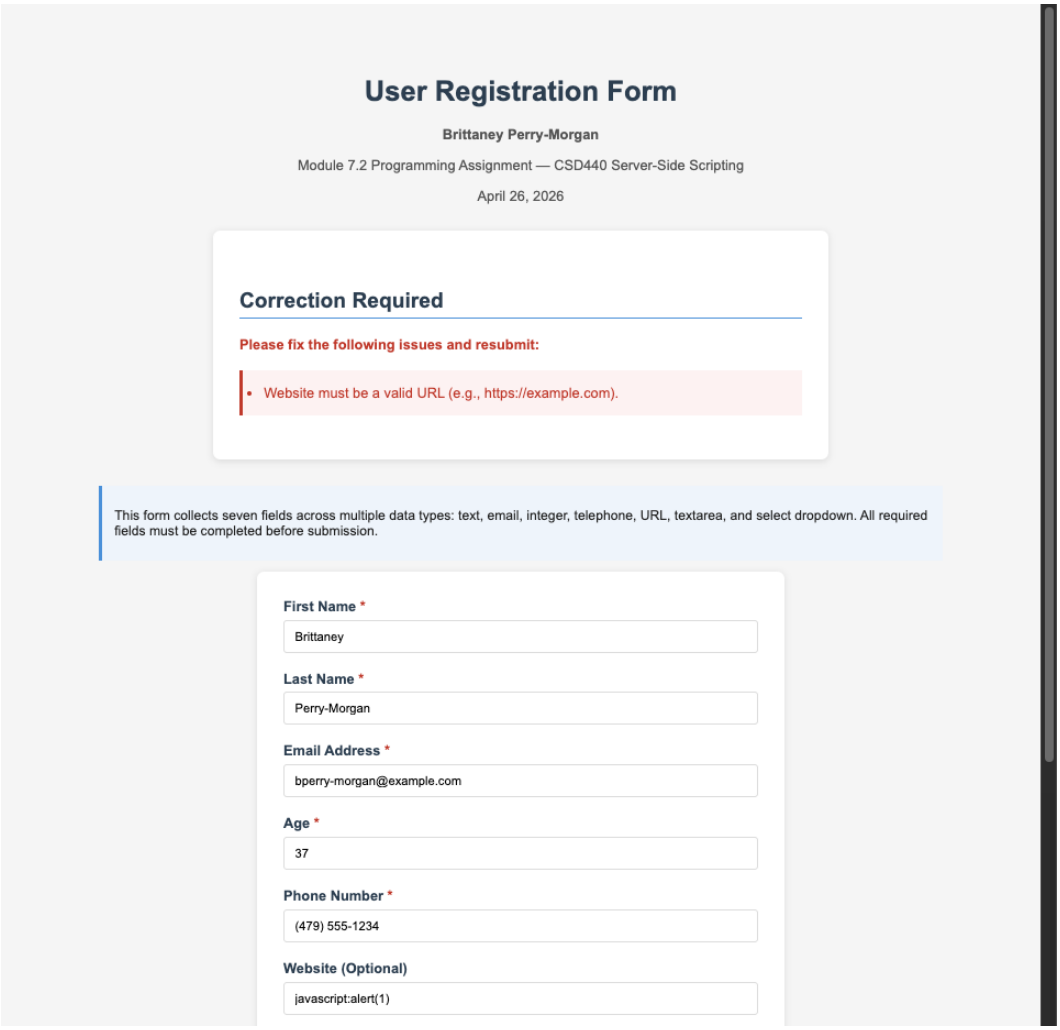
Field	Value
First Name	Brittaney
Last Name	Perry-Morgan
Email	bperry-morgan@example.com
Age	37
Phone	(479) 555-1234
Website	ftp://badprotocol.com
Bio	Testing invalid URL.
Contact Preference	Email

Submit Another Registration

08 Invalid Url Ftp Protocol

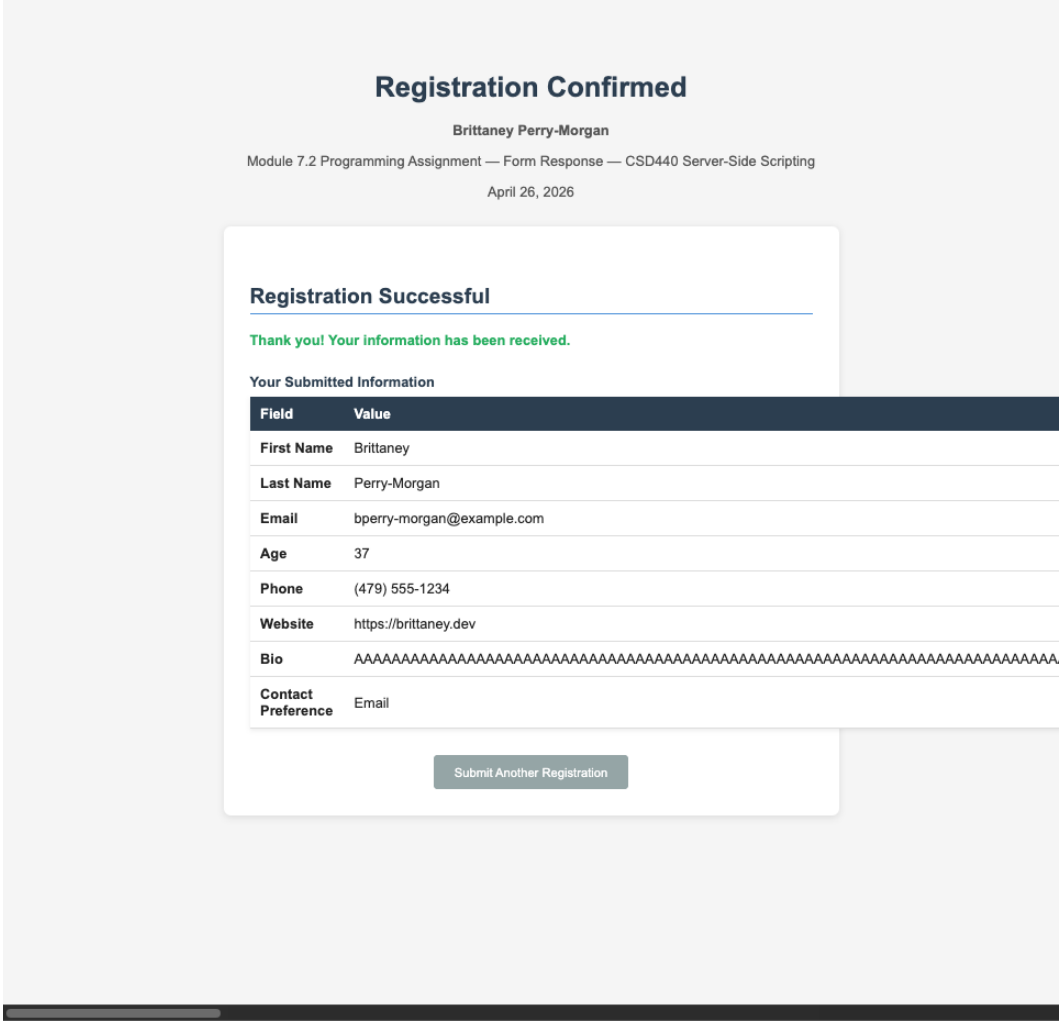


08 Invalid Url Incomplete Url



08 Invalid URL JavaScript Protocol

Test 9: Boundary and Length Testing



Registration Confirmed

Brittaney Perry-Morgan

Module 7.2 Programming Assignment — Form Response — CSD440 Server-Side Scripting

April 26, 2026

Registration Successful

Thank you! Your information has been received.

Your Submitted Information

Field	Value
First Name	Brittaney
Last Name	Perry-Morgan
Email	bperry-morgan@example.com
Age	37
Phone	(479) 555-1234
Website	https://brittaney.dev
Bio	AAAA

Registration Confirmed

Brittaney Perry-Morgan

Module 7.2 Programming Assignment — Form Response — CSD440 Server-Side Scripting

April 26, 2026

Registration Successful

Thank you! Your information has been received.

Your Submitted Information

Field	Value
First Name	Brittaney
Last Name	Perry-Morgan
Email	bperry-morgan@example.com
Age	18
Phone	(479) 555-1234
Website	https://brittaney.dev
Bio	Testing age boundary value: 18
Contact Preference	Email

Submit Another Registration

09 Age Boundary 18

Registration Confirmed

Brittaney Perry-Morgan

Module 7.2 Programming Assignment — Form Response — CSD440 Server-Side Scripting

April 26, 2026

Registration Successful

Thank you! Your information has been received.

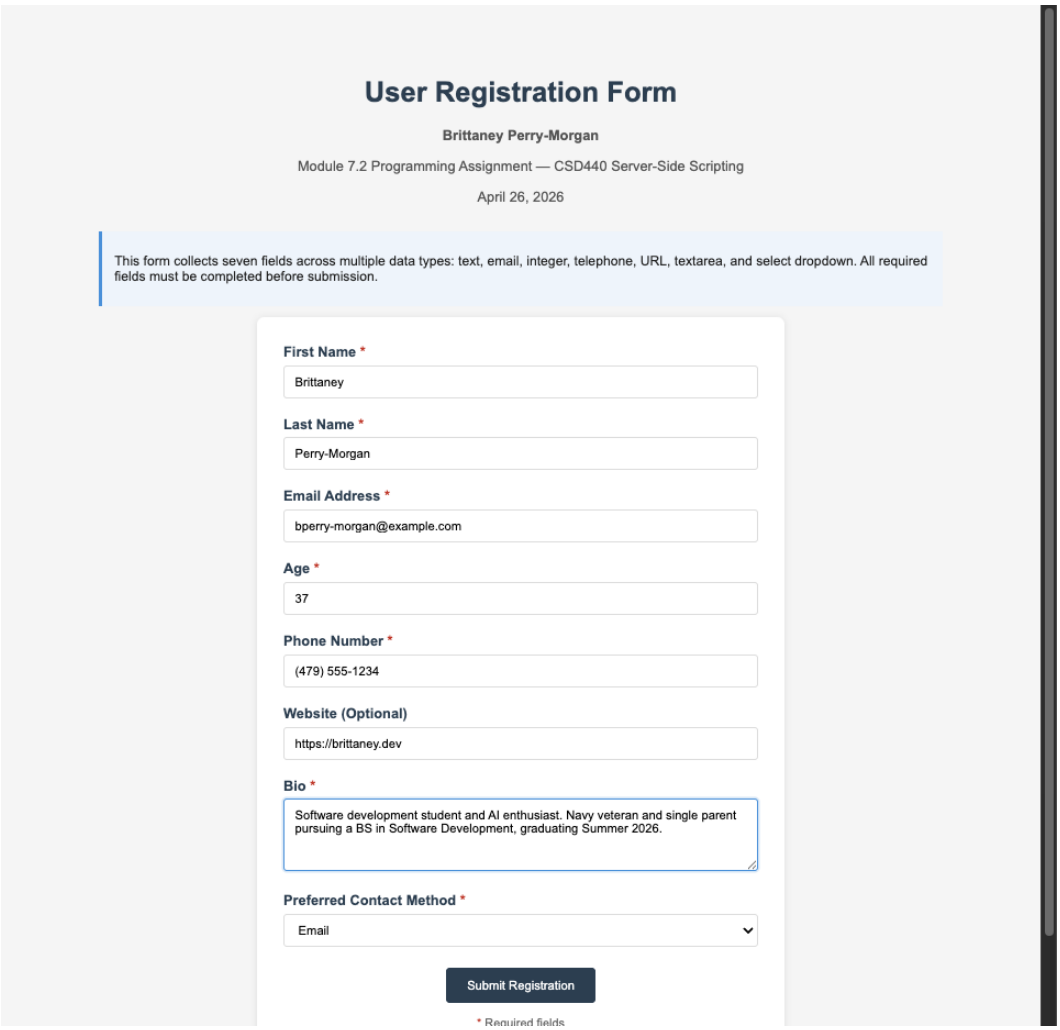
Your Submitted Information

Field	Value
First Name	Brittaney
Last Name	Perry-Morgan
Email	bperry-morgan@example.com
Age	120
Phone	(479) 555-1234
Website	https://brittaney.dev
Bio	Testing age boundary value: 120
Contact Preference	Email

Submit Another Registration

09 Age Boundary 120

Test 10: Valid Submission — Success Page



10 Valid Submission Filled

Registration Confirmed

Brittaney Perry-Morgan

Module 7.2 Programming Assignment — Form Response — CSD440 Server-Side Scripting

April 26, 2026

Registration Successful

Thank you! Your information has been received.

Your Submitted Information

Field	Value
First Name	Brittaney
Last Name	Perry-Morgan
Email	bperry-morgan@example.com
Age	37
Phone	(479) 555-1234
Website	https://brittaney.dev
Bio	Software development student and AI enthusiast. Navy veteran and single parent pursuing a BS in Software Development, graduating Summer 2026.
Contact Preference	Email

Submit Another Registration

10 Success Page